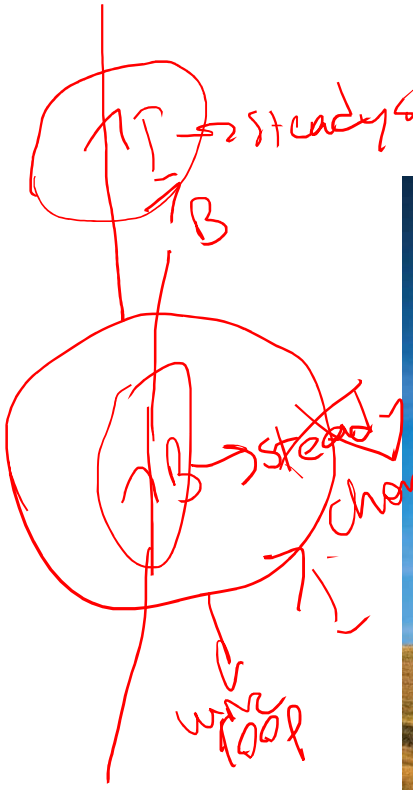
Storyline

Chapter 30: Faraday's Law



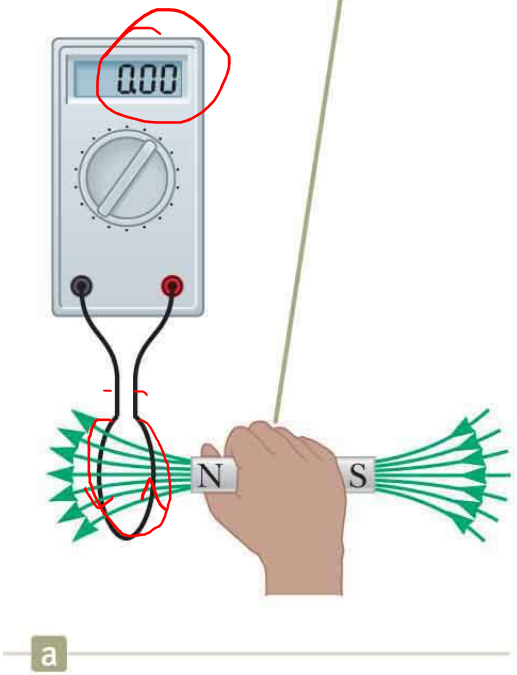


Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

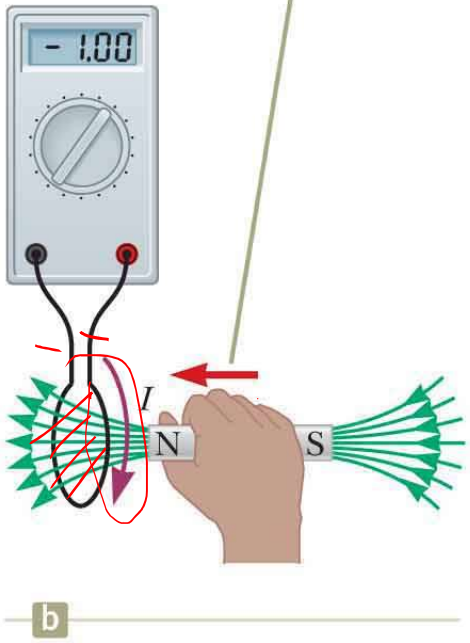
BROOKS/COLE
CENGAGE Learning

Faraday's Law of Induction

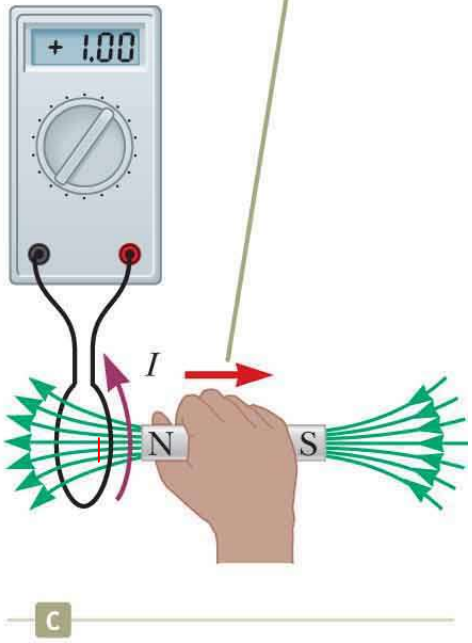
When a magnet is held stationary near a loop of wire connected to a sensitive ammeter, there is no induced current in the loop, even when the magnet is inside the loop.



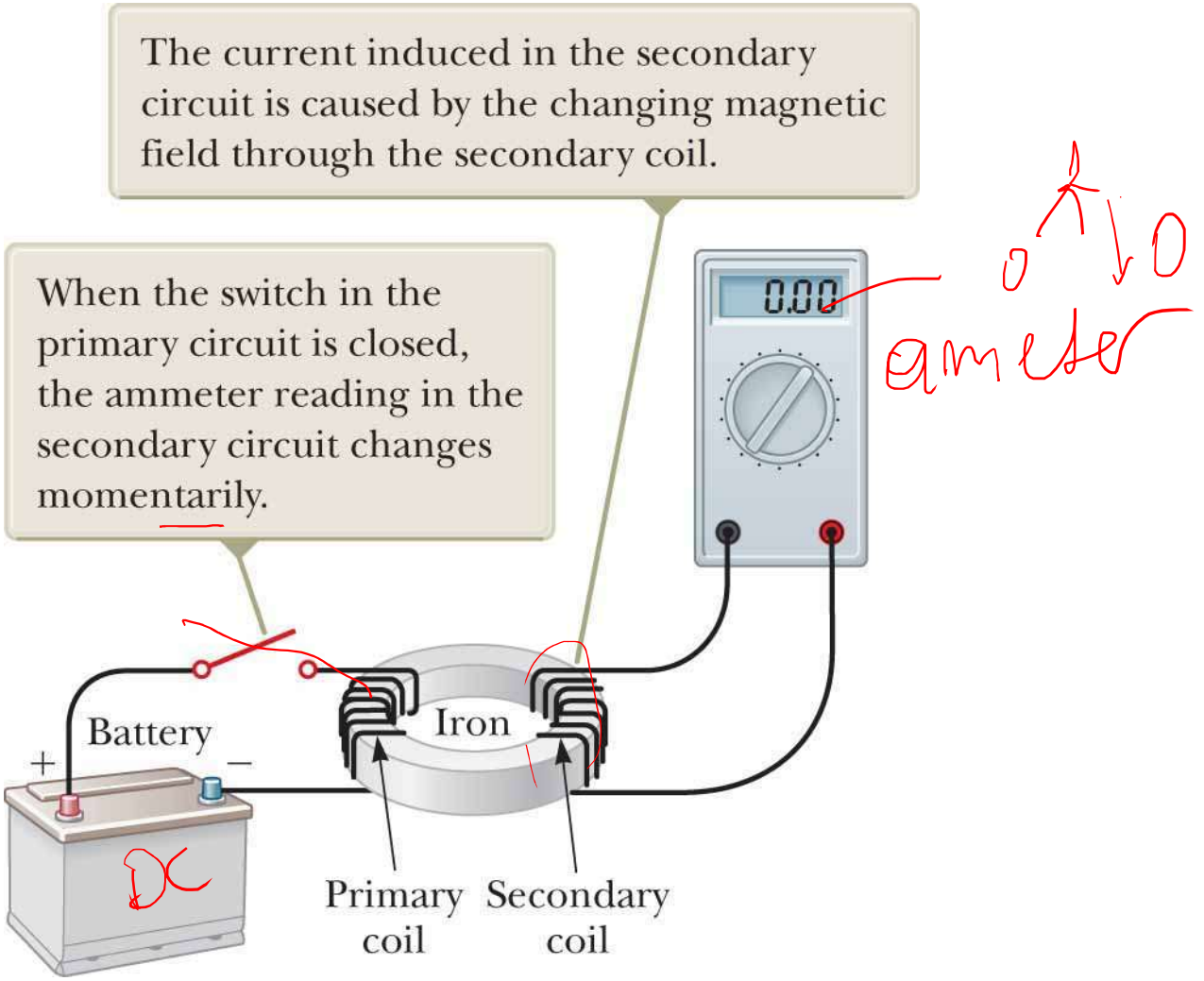
When the magnet is moved toward the loop of wire, the ammeter shows that a current is induced in the loop.



When the magnet is moved away from the loop, the ammeter shows that the induced current is opposite that shown in part b.



Faraday's Law of Induction



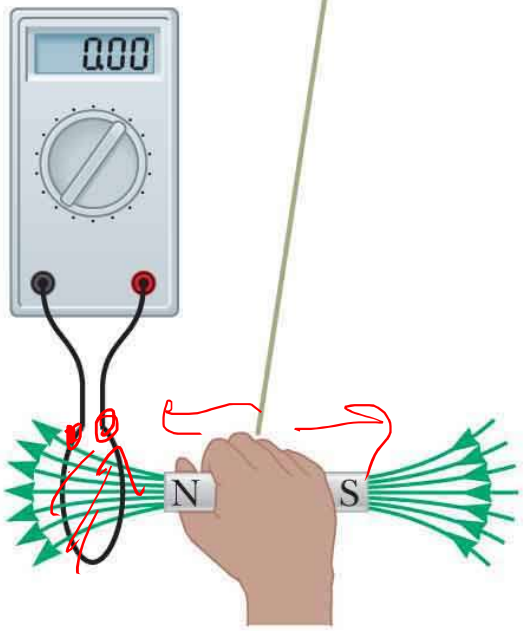
Faraday's Law of Induction

When a magnet is held stationary near a loop of wire connected to a sensitive ammeter, there is no induced current in the loop, even when the magnet is inside the loop.

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

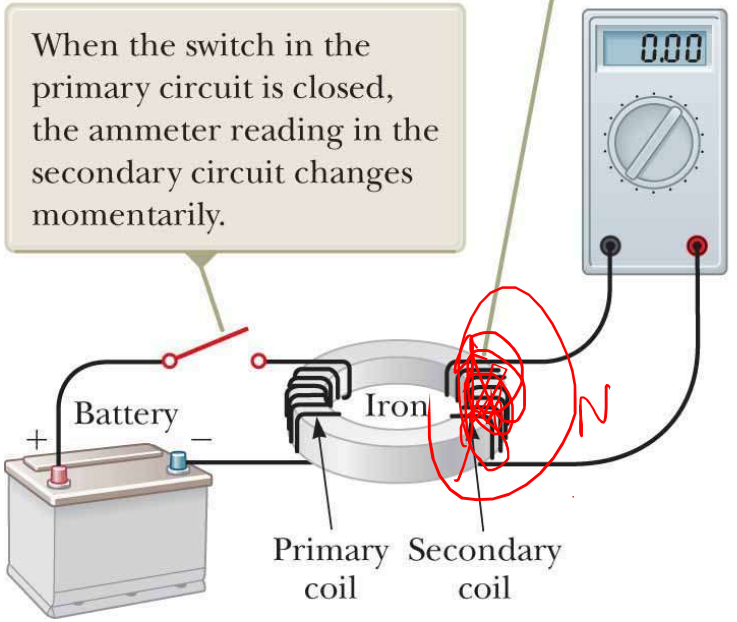
$$\mathcal{E} = -N\frac{d\Phi_B}{dt}$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

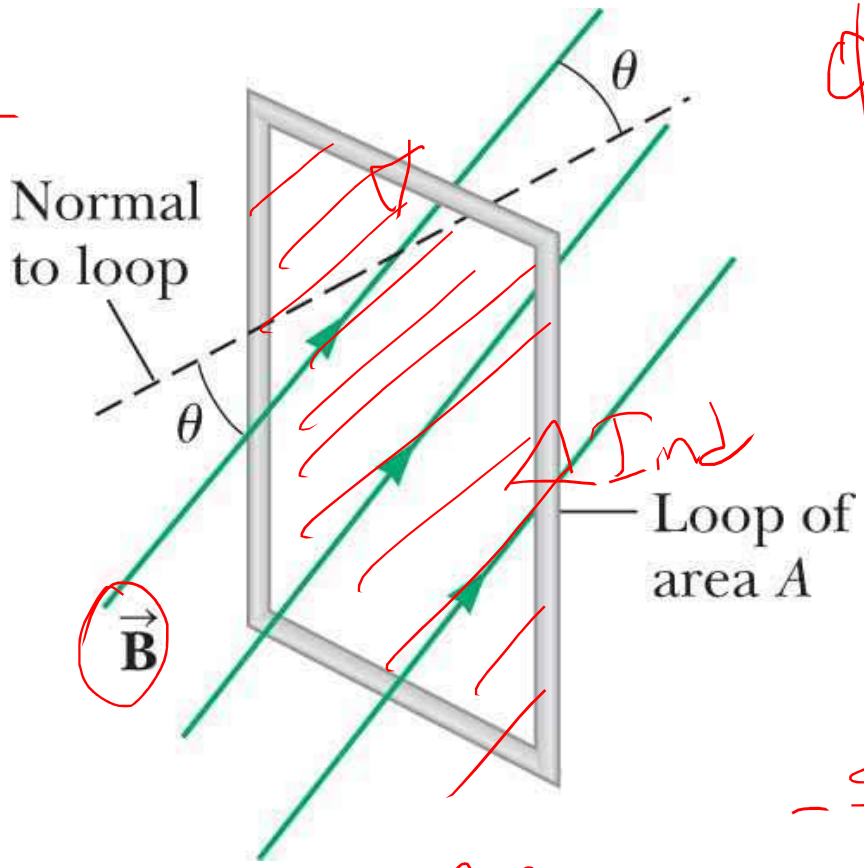


The current induced in the secondary circuit is caused by the changing magnetic field through the secondary coil.

When the switch in the primary circuit is closed, the ammeter reading in the secondary circuit changes momentarily.



Faraday's Law of Induction



$$\varepsilon = -\frac{d}{dt}(BA\cos\theta)$$

Quick Quiz 30.1

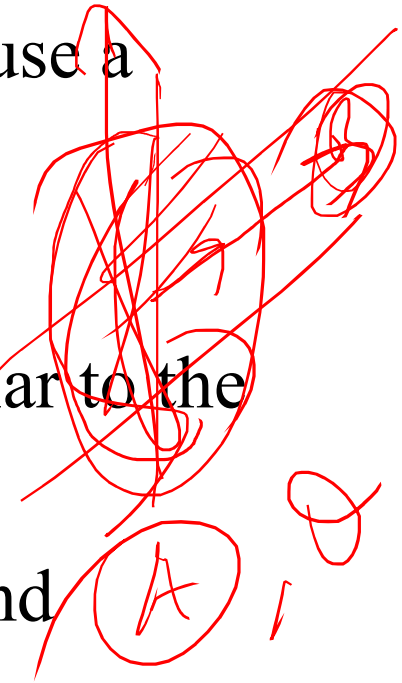
A circular loop of wire is held in a uniform magnetic field, with the plane of the loop perpendicular to the field lines. Which of the following will not cause a current to be induced in the loop?

(a) crushing the loop

(b) rotating the loop about an axis perpendicular to the field lines

(c) keeping the orientation of the loop fixed and moving it along the field lines

(d) pulling the loop out of the field



Quick Quiz 30.1

A circular loop of wire is held in a uniform magnetic field, with the plane of the loop perpendicular to the field lines. Which of the following will not cause a current to be induced in the loop?

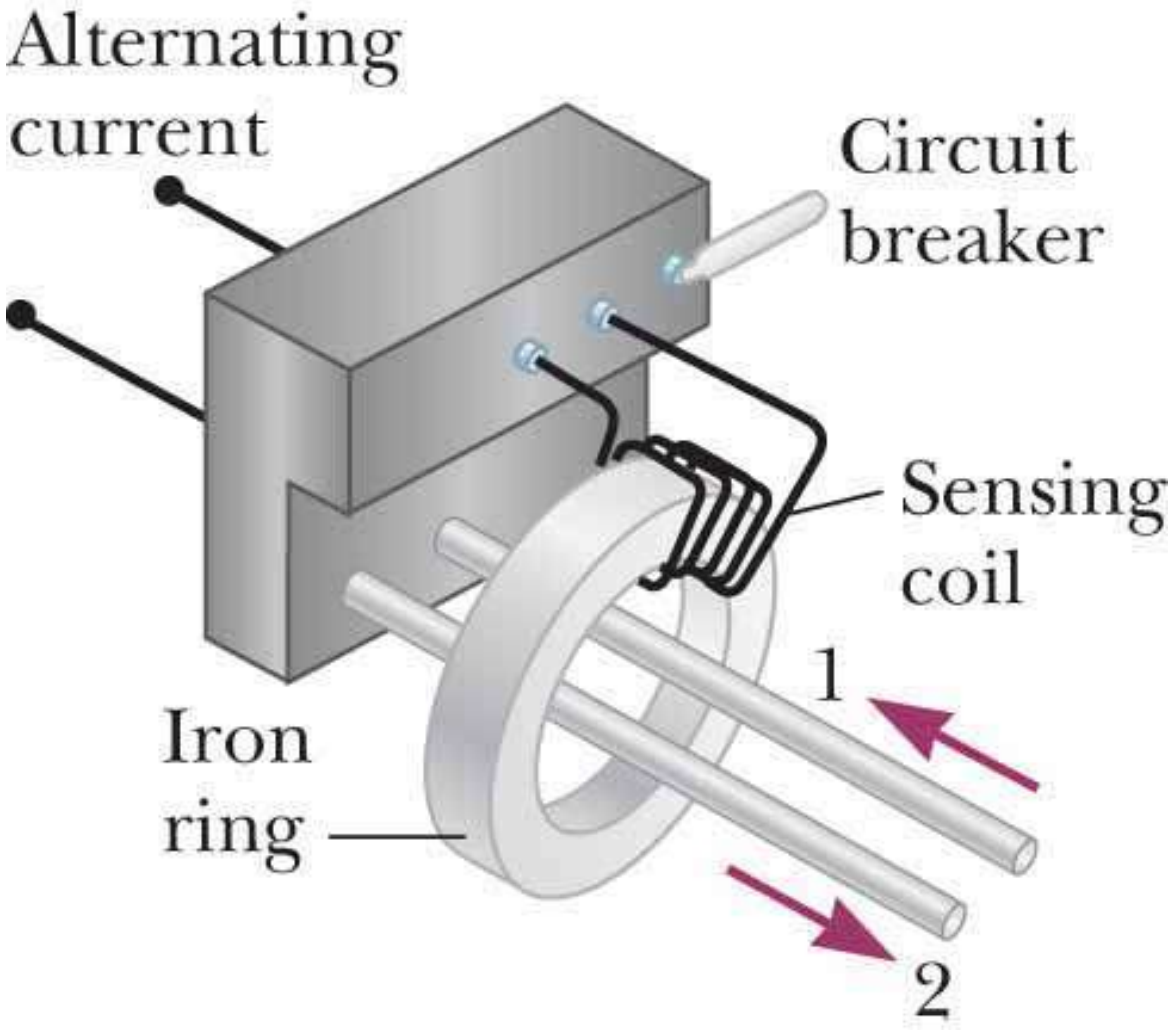
(a) crushing the loop

(b) rotating the loop about an axis perpendicular to the field lines

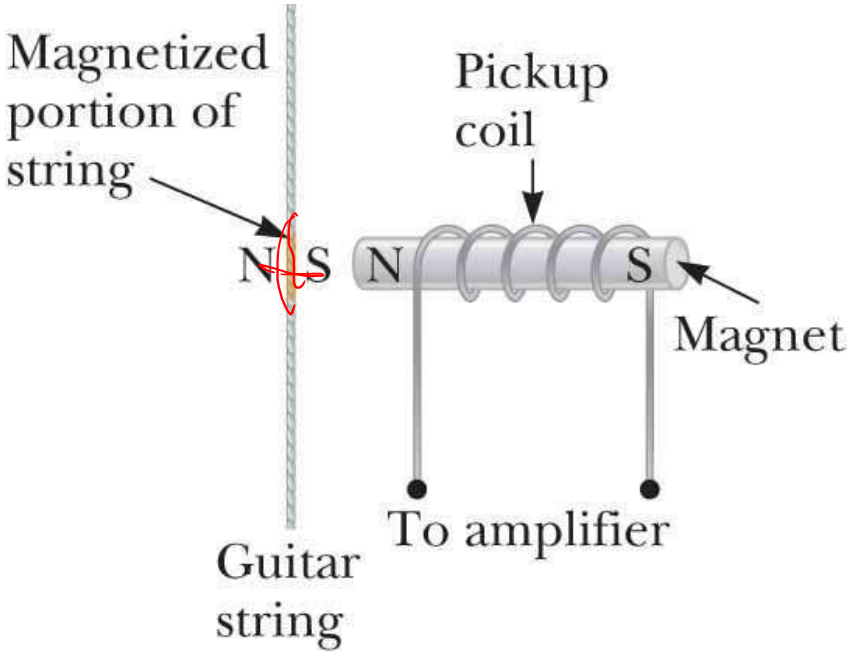
(c) keeping the orientation of the loop fixed and moving it along the field lines

(d) pulling the loop out of the field

Some Applications of Faraday's Law



Some Applications of Faraday's Law



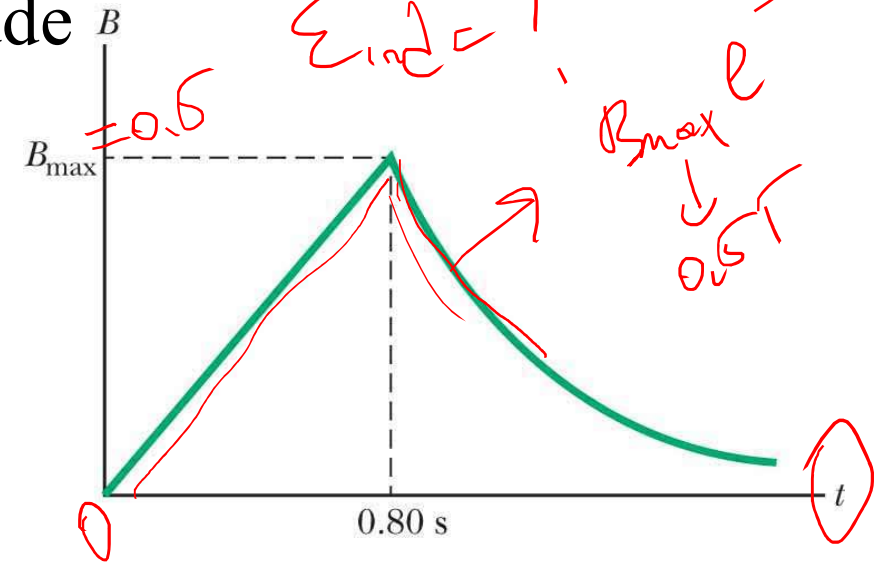
a



b

Example 30.1: Inducing an emf in a Coil

A coil consists of 200 turns of wire. Each turn is a square of side $d = 18$ cm, and a uniform magnetic field directed perpendicular to the plane of the coil is turned on. The figure shows the behavior of the magnitude of the magnetic field with time. From $t = 0$ to $t = 0.80$ s, the field changes linearly from 0 to 0.50 T. After $t = 0.80$ s, the magnitude of the field decays in time according to the expression $B = B_{\max} e^{-at}$, where a is some constant and $B_{\max} = 0.50$ T.



Example 30.1:
Inducing an emf in a Coil

(A) What is the magnitude of the induced emf in the coil between $t = 0$ and $t = 0.80$ s?

$$|\varepsilon| = N\frac{\Delta\Phi_B}{\Delta t} = N\frac{\Delta(BA)}{\Delta t} = NA\frac{\Delta B}{\Delta t} = Nd^2\frac{B_f - B_i}{\Delta t}$$

$$|\varepsilon| = (200)(0.18\ \text{m})^2\frac{(0.50\ \text{T} - 0)}{0.80\ \text{s}} = \boxed{4.0\ \text{V}}$$

Example 30.1: Inducing an emf in a Coil

(B) What is the magnitude of the induced emf in the coil after $t = 0.80$ s?

$$\varepsilon = -N\frac{d\Phi_B}{dt} = N\frac{d\left(AB_{\max}e^{-at}\right)}{dt}$$

$$= -NAB_{\max}\frac{d}{dt}e^{-at} = \boxed{aNd^2B_{\max}e^{-at}}$$

$$\varepsilon = a(200)(0.18\text{ m})^2(0.50\text{ T})e^{-at} = 3.2ae^{-at}$$

Example 30.1: Inducing an emf in a Coil

What if you were asked to find the magnitude of the induced current in the coil while the field is changing during the first 0.80 s? Can you answer that question?

* ends of the coil connected to an external circuit and resistance of the coil is 2Ω

$$I = \frac{|\varepsilon|}{R} = \frac{4.0\ \text{V}}{2.0\ \Omega} = 2.0\ \text{A}$$

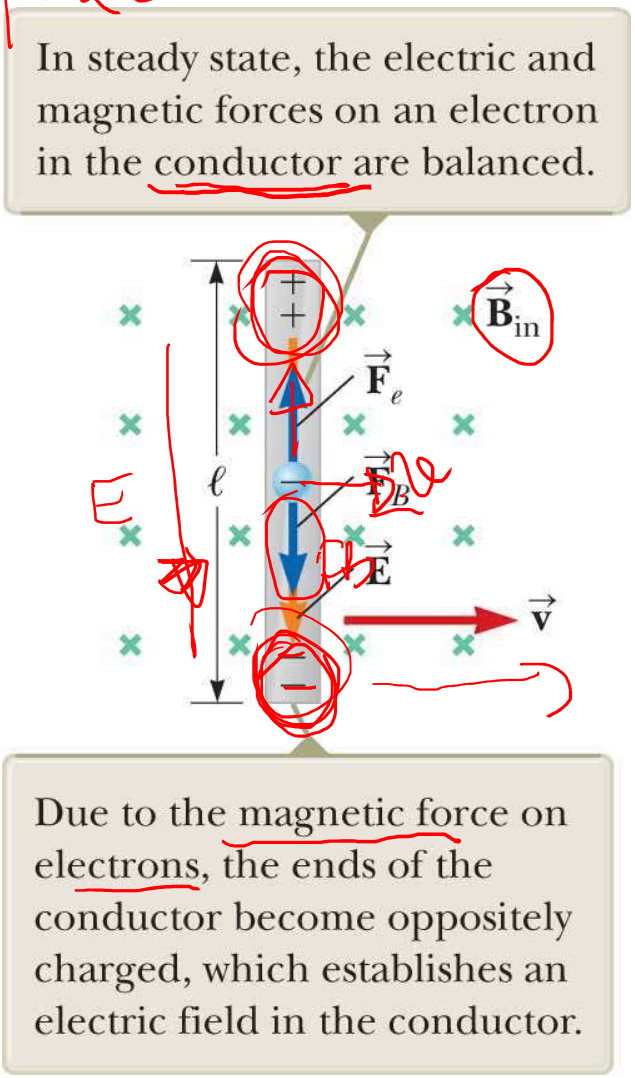
* if the end of the coil are not connected to a circuit, the current is zero.

Motional emf

$$\sum F = 0 \Rightarrow qE - qvB = 0$$

$$\Rightarrow E = vB$$

$$\mathcal{E}_{ind} = -\Delta V = E\ell = B\ell v$$

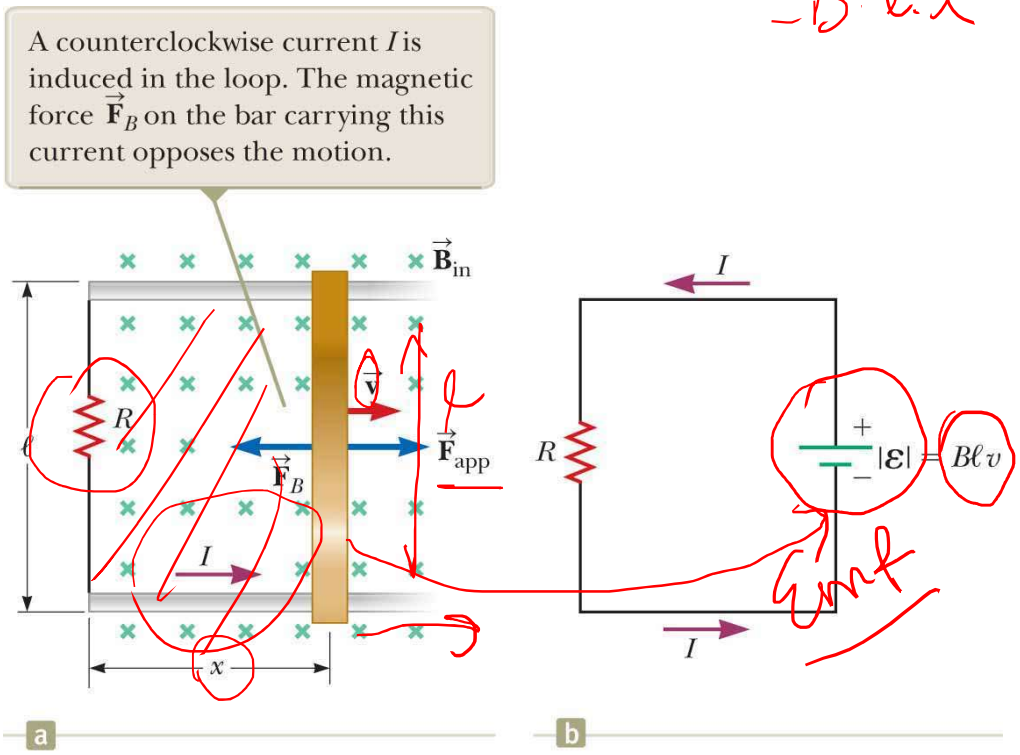


Motional emf

$$\Phi_B = B\ell x$$

$$\varepsilon = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}(B\ell x) = -B\ell\frac{dx}{dt} \Rightarrow \varepsilon = -B\ell v$$

$$I = \frac{|\varepsilon|}{R} = \frac{B\ell v}{R}$$

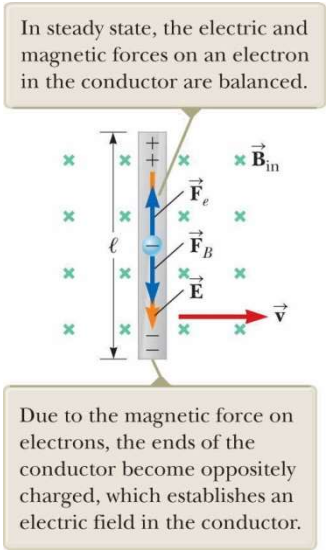


Motional emf: Energy Approach

$$|\varepsilon| = B\ell v$$

$$0 = W_{\text{app}} + T_{\text{ET}}$$

$$\frac{dW_{\text{app}}}{dt} = -\frac{dT_{\text{ET}}}{dt} \Rightarrow P_{\text{app}} = -P_{\text{elec}}$$

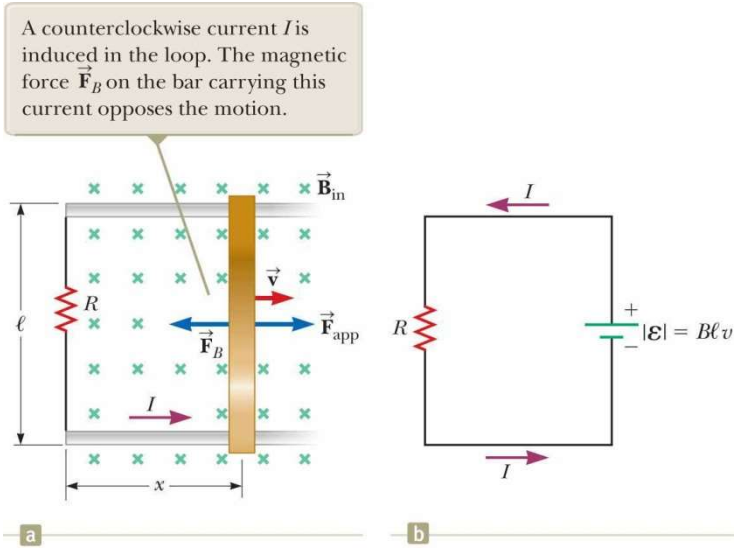


$$P_{\text{app}} = F_{\text{app}} v = F_B v$$

$$= (I\ell B)v = I(B\ell v)$$

$$= I\varepsilon = I(IR)$$

$$= I^2 R = -P_{\text{elec}}$$



Quick Quiz 30.2

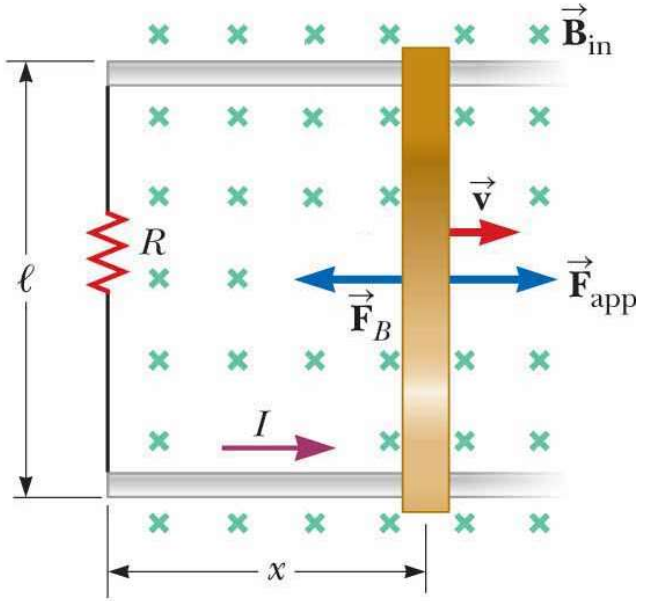
In the figure, a given applied force of magnitude F results in a constant speed v and a power input P. Imagine that the force is increased so that the constant speed of the bar is doubled to $2v$. Under these conditions, what are the new force and the new power input?

(a) $2F$ and $2P$

(b) $4F$ and $2P$

(c) $2F$ and $4P$

(d) $4F$ and $4P$



Quick Quiz 30.2

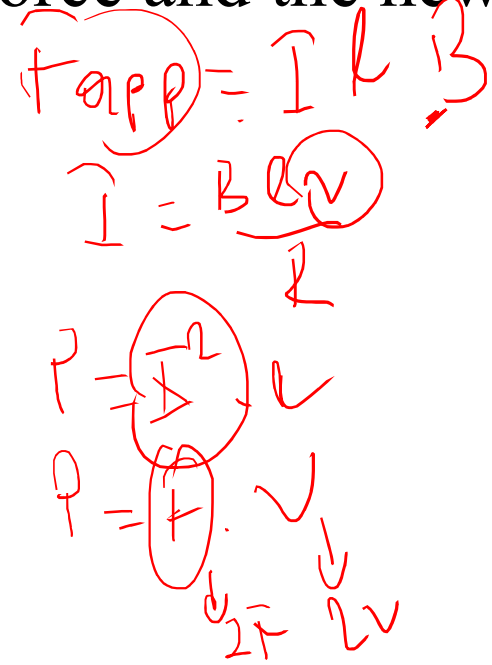
In the figure, a given applied force of magnitude F results in a constant speed v and a power input P. Imagine that the force is increased so that the constant speed of the bar is doubled to $2v$. Under these conditions, what are the new force and the new power input?

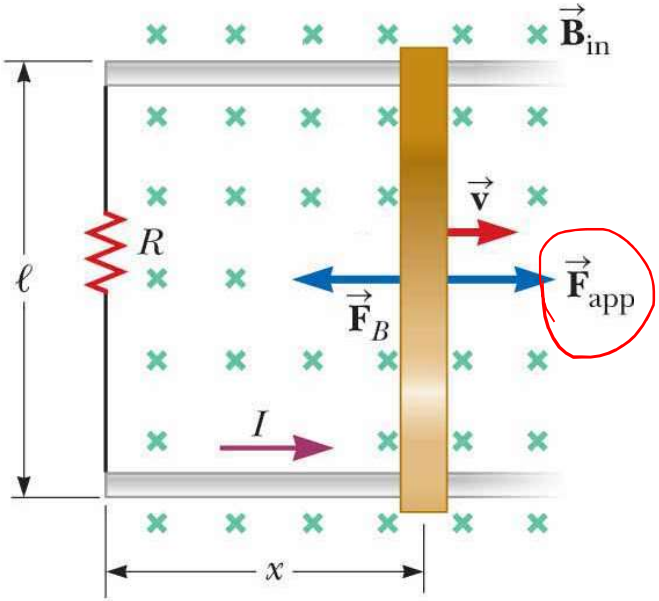
(a) $2F$ and $2P$

(b) $4F$ and $2P$

(c) $2F$ and $4P$

(d) $4F$ and $4P$

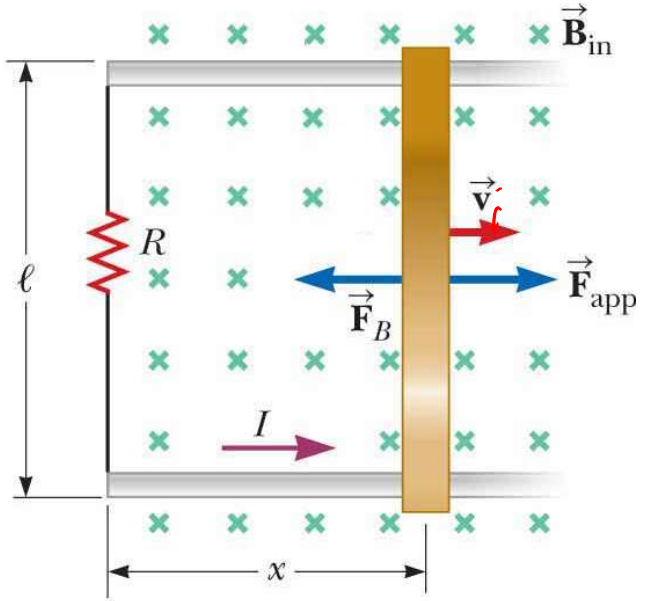




Example 30.2:
Magnetic Force Acting on a Sliding Bar

The conducting bar illustrated in the figure moves on two frictionless, parallel rails in the presence of a uniform magnetic field directed into the page. The bar has mass m, and its length is ℓ. The bar is given an initial velocity $\vec{\mathbf{v}}_i$ to the right and is released at $t = 0$.

(A) Using Newton's laws, find the speed of the bar as a function of time after it is released.



Example 30.2:
Magnetic Force Acting on a Sliding Bar

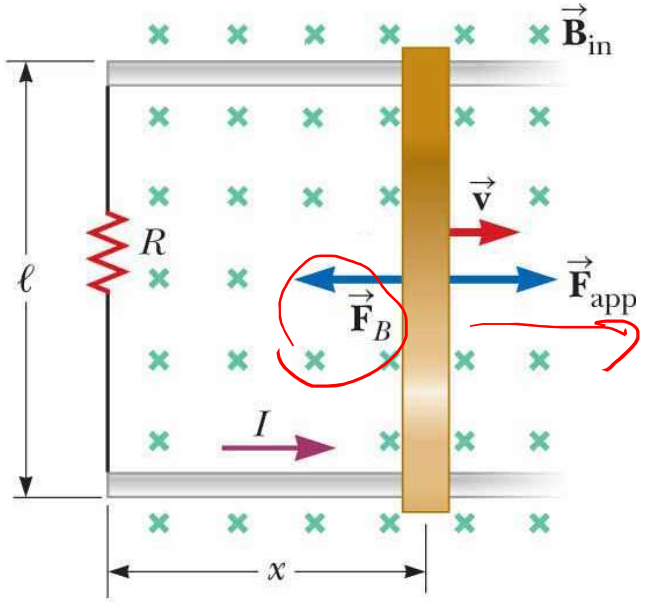
$$F_x = ma \Rightarrow -I\ell B = m\frac{dv}{dt}$$

$$m\frac{dv}{dt} = -\left(\frac{B\ell v}{R}\right)\ell B = -\frac{B^2\ell^2}{R}v \quad \Rightarrow \frac{dv}{v} = -\left(\frac{B^2\ell^2}{mR}\right)dt$$

$$\int_{v_i}^{v}\frac{dv}{v} = -\frac{B^2\ell^2}{mR}\int_0^t dt$$

$$\ln\left(\frac{v}{v_i}\right) = -\left(\frac{B^2\ell^2}{mR}\right)t$$

$$v = v_i e^{-t/\tau}, \quad \tau = \frac{mR}{B^2\ell^2}$$



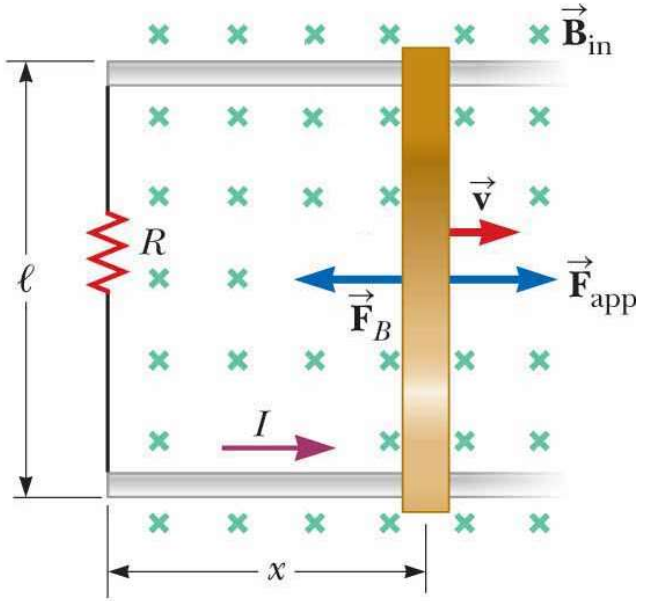
Example 30.2:
Magnetic Force Acting on a Sliding Bar

(B) Show that the same result is found by using an energy approach.

$$\frac{dK}{dt} = \frac{dT_{\text{ET}}}{dt} = P_{\text{elec}} = -I^2 R$$

$$\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = -\left(\frac{B\ell v}{R}\right)^2 R$$

$$\Rightarrow mv\frac{dv}{dt} = -\frac{\left(B\ell v\right)^2}{R}$$



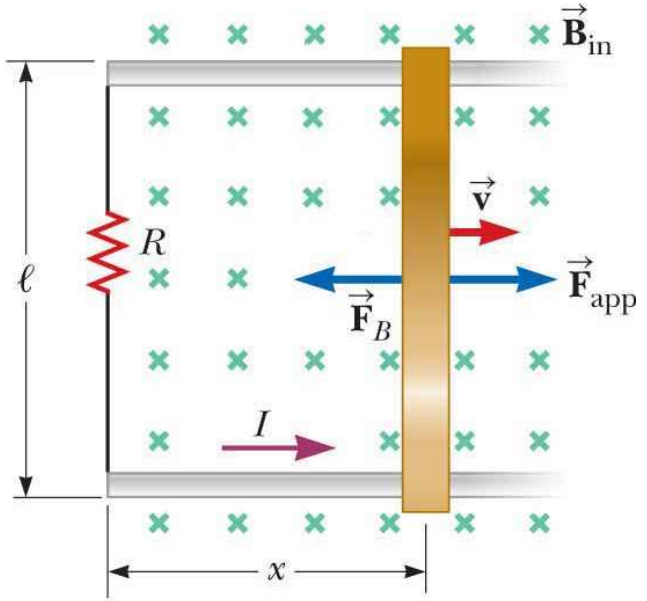
Example 30.2:
Magnetic Force Acting on a Sliding Bar

Suppose you wished to increase the distance through which the bar moves between the time it is initially projected and the time it essentially comes to rest. You can do so by changing one of three variables—v_i, R, or B—by a factor of 2 or ½. Which variable should you change to maximize the distance, and would you double it or halve it?

$$v = dx/dt = v_i e^{-t/\tau}$$

$$x = \int_0^\infty v_i e^{-t/\tau} dt = -v_i \tau e^{-t/\tau} \Big|_0^\infty$$

$$= -v_i \tau (0 - 1) = v_i \tau = v_i \left(\frac{mR}{B^2 \ell^2} \right)$$



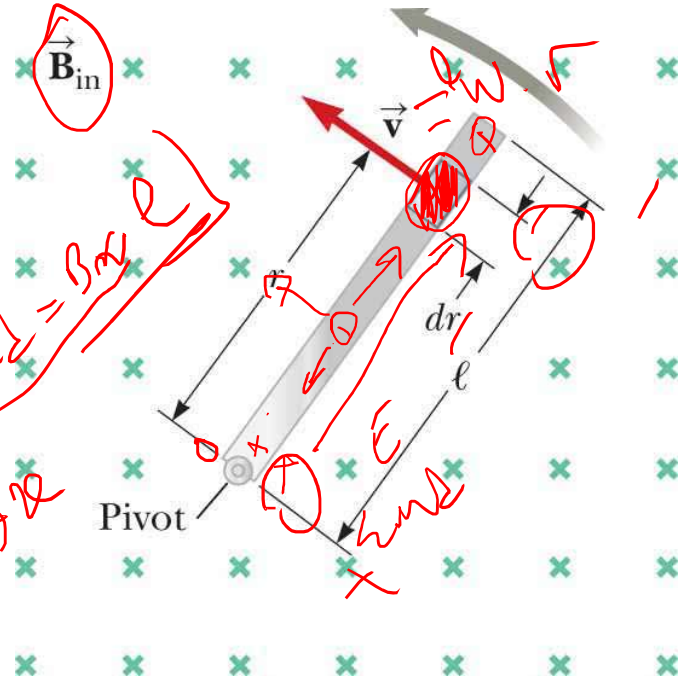
Example 30.3:
Motional emf Induced in a Rotating Bar

A conducting bar of length ℓ rotates with a constant angular speed v about a pivot at one end. A uniform magnetic field $\vec{\mathbf{B}}$ is directed perpendicular to the plane of rotation as shown in the figure. Find the motional emf induced between the ends of the bar.

$$d\varepsilon = Bv\,dr \Rightarrow \varepsilon = \int Bv\,dr$$

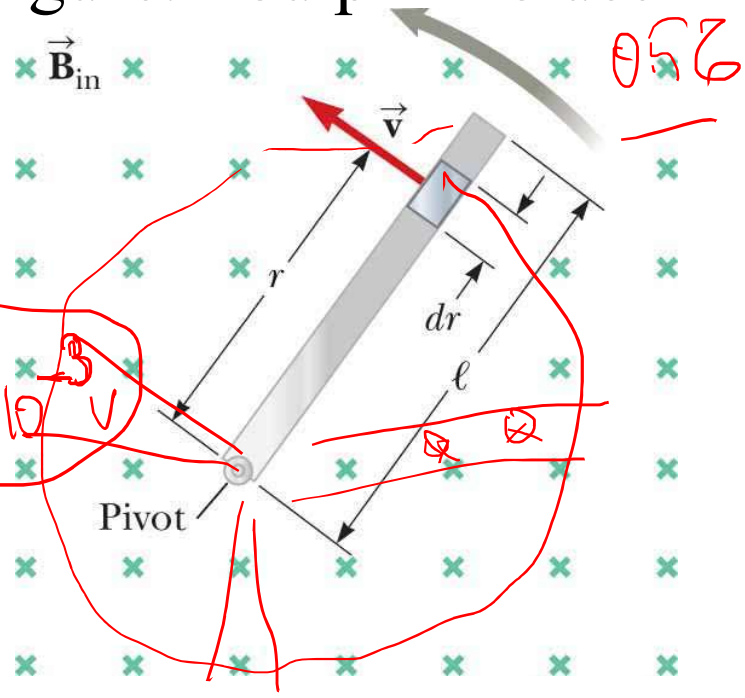
$$\varepsilon = B\int v\,dr = B\omega\int_0^{\ell} r\,dr$$

$$= \boxed{\frac{1}{2}B\omega\ell^2}$$

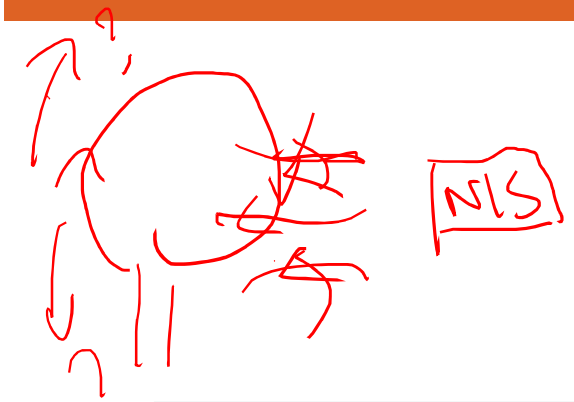


Example 30.3:
Motional emf Induced in a Rotating Bar

Suppose, after reading through this example, you come up with a brilliant idea. A Ferris wheel has radial metallic spokes between the hub and the circular rim. These spokes move in the magnetic field of the Earth, so each spoke acts like the bar in the figure. You plan to use the emf generated by the rotation of the Ferris wheel to power the incandescent lightbulbs on the wheel. Will this idea work?



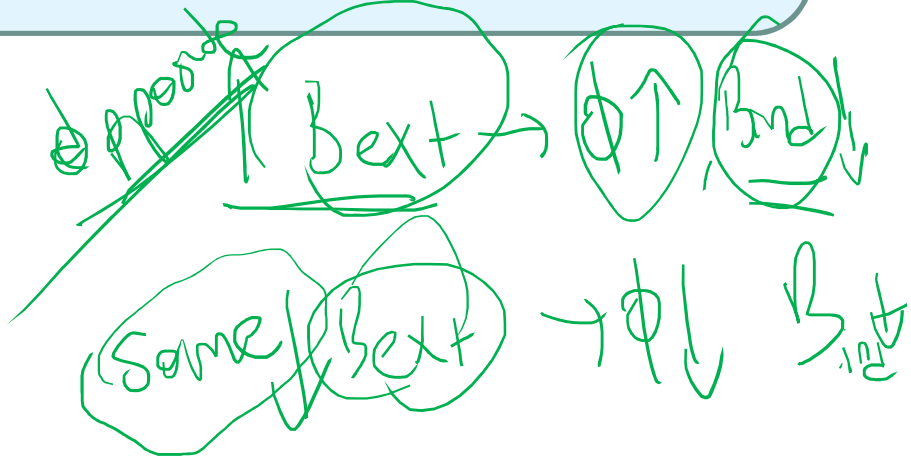
Lenz's Law



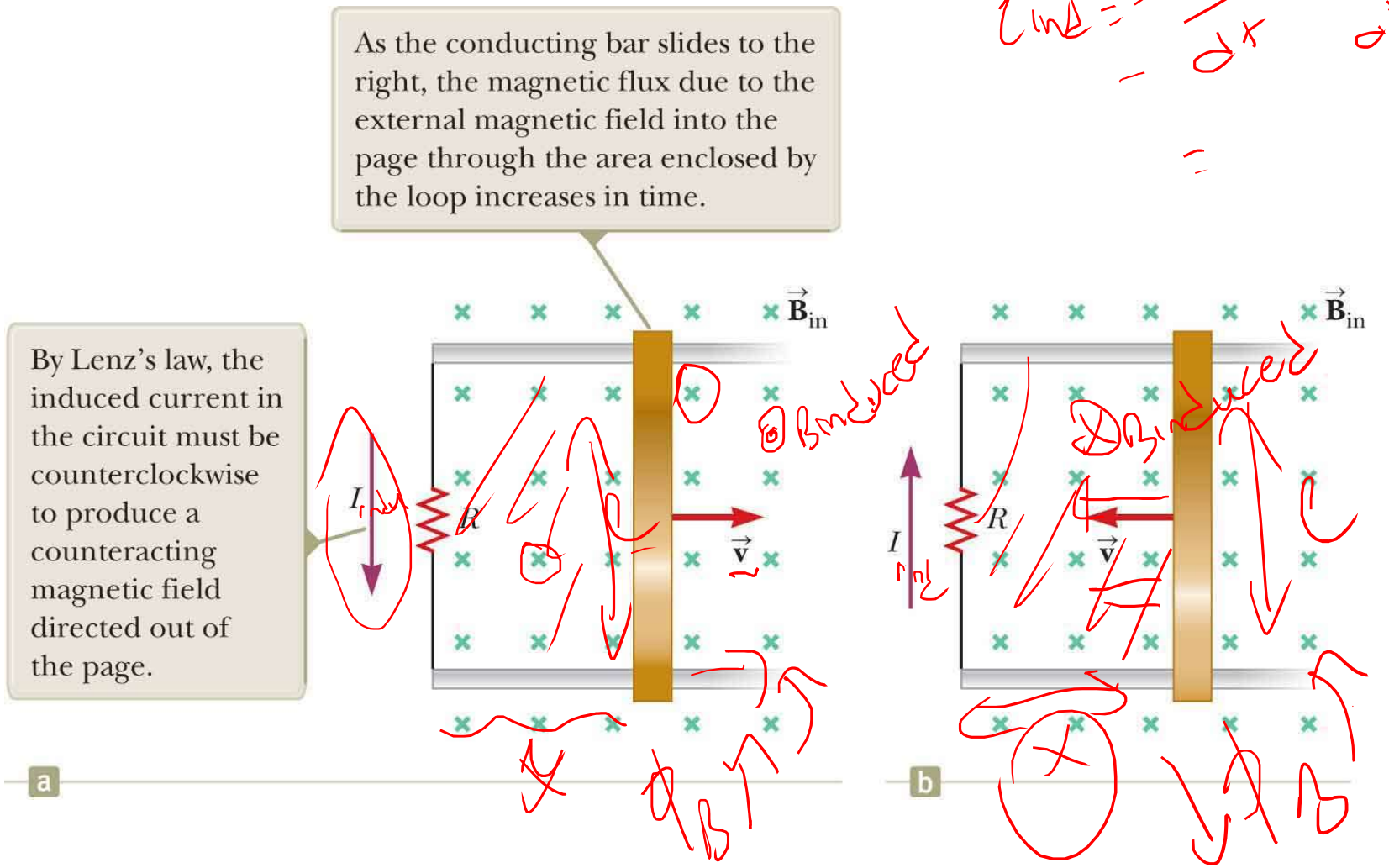
$$\varepsilon = -\frac{d\Phi_B}{dt}$$



The induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop.



Lenz's Law


As the conducting bar slides to the right, the magnetic flux due to the external magnetic field into the page through the area enclosed by the loop increases in time.
By Lenz's law, the induced current in the circuit must be counterclockwise to produce a counteracting magnetic field directed out of the page.
$\vec{B}_{in}$
R
$\vec{v}$
I
a
b

Quick Quiz 30.3

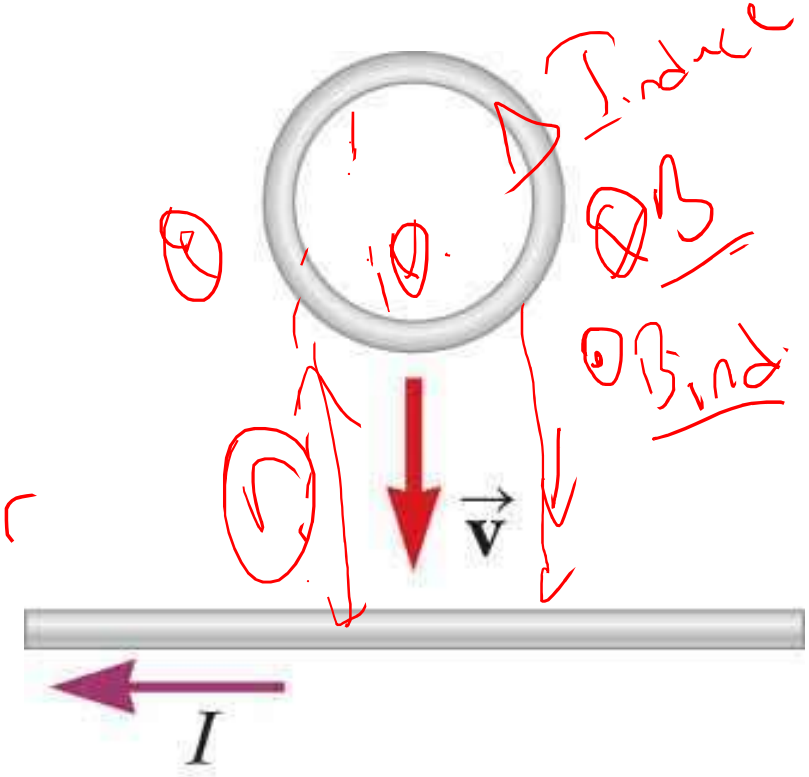
The figure shows a circular loop of wire falling toward a wire carrying a current to the left. What is the direction of the induced current in the loop of wire?

(a) clockwise

(b) counterclockwise

(c) zero

(d) impossible to determine



Quick Quiz 30.3

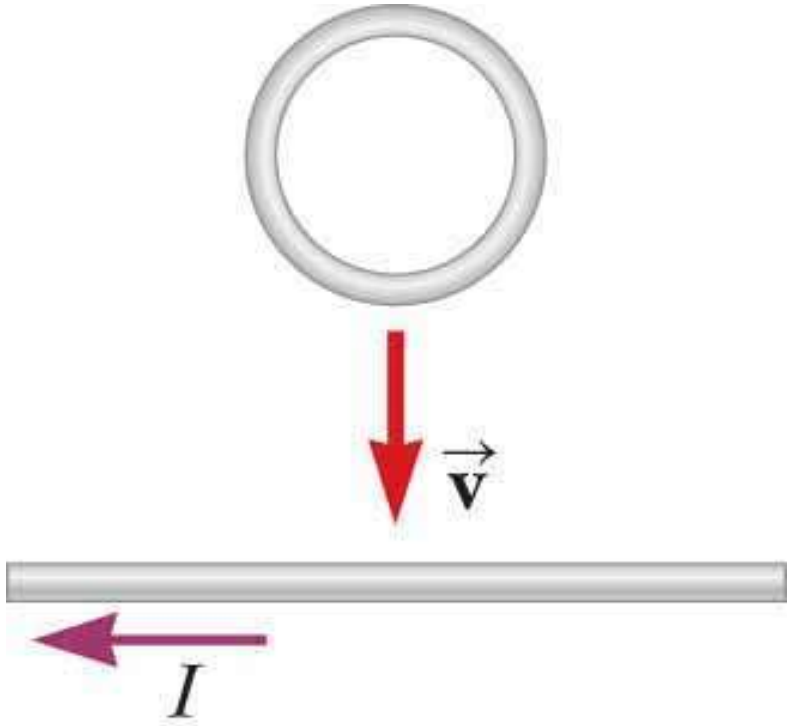
The figure shows a circular loop of wire falling toward a wire carrying a current to the left. What is the direction of the induced current in the loop of wire?

(a) clockwise

(b) counterclockwise

(c) zero

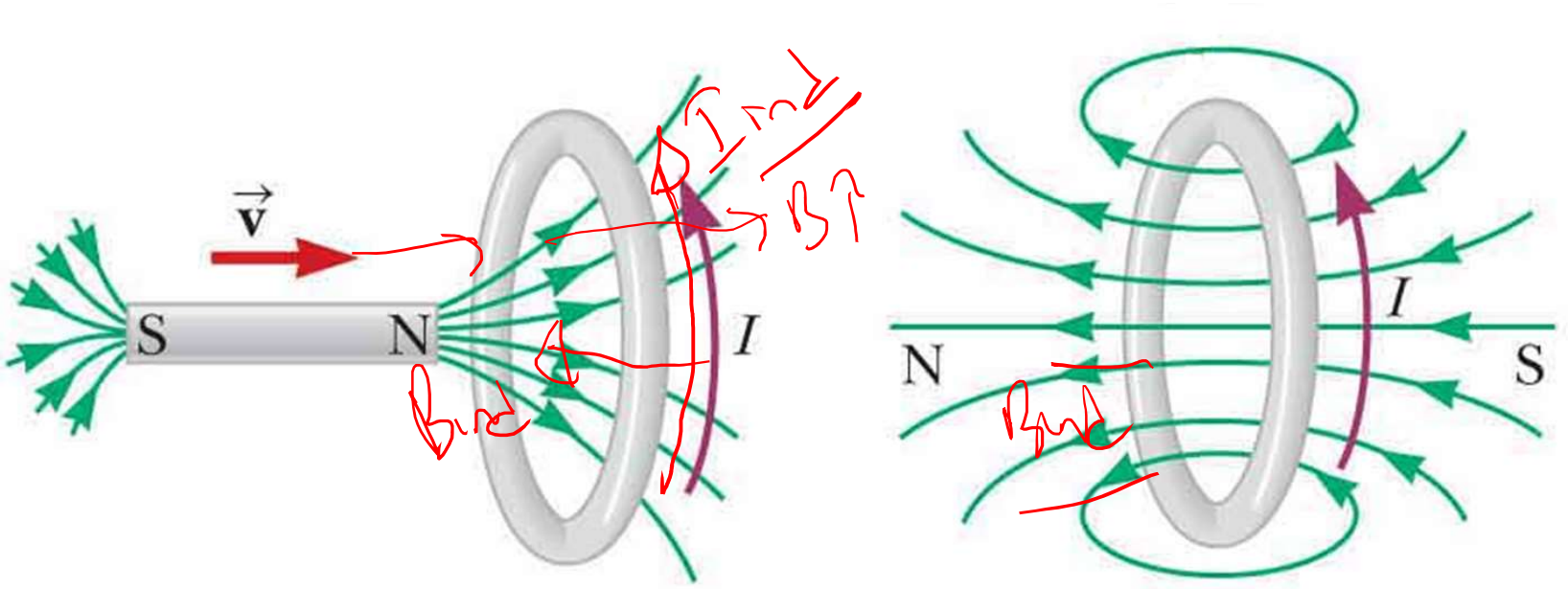
(d) impossible to determine



Conceptual Example 30.4: Application of Lenz's Law

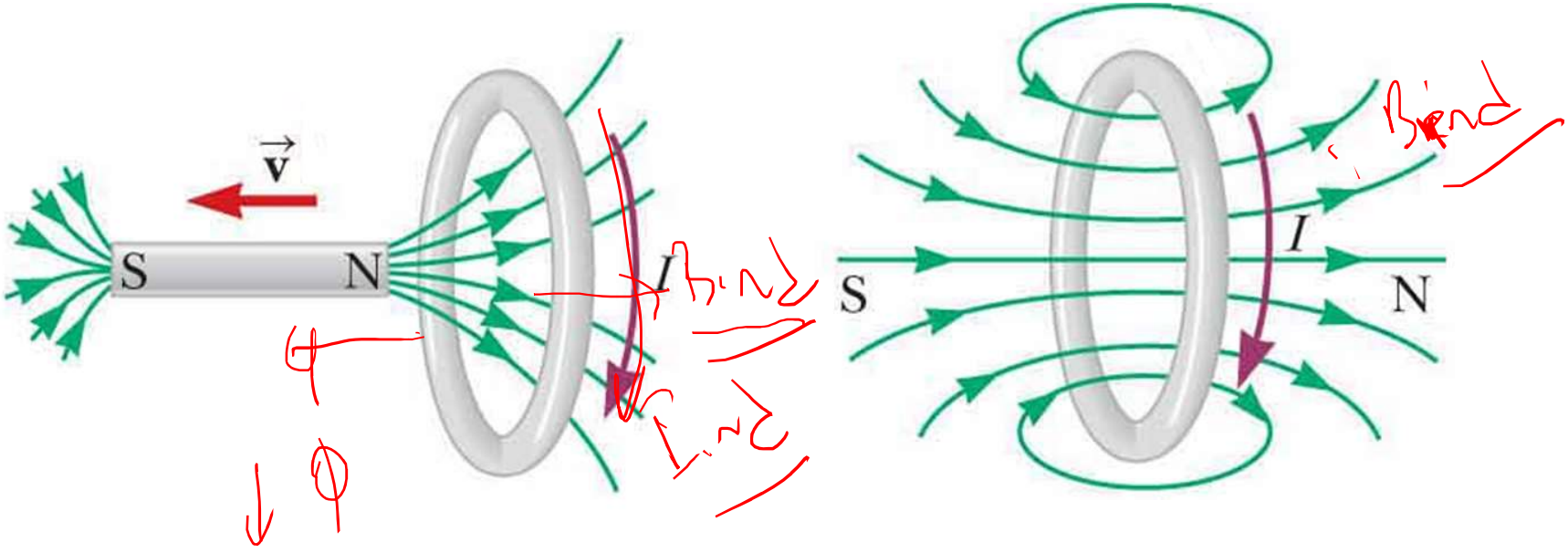
A magnet is placed near a metal loop as shown in the figure.

(A) Find the direction of the induced current in the loop when the magnet is pushed toward the loop.



Conceptual Example 30.4: Application of Lenz's Law

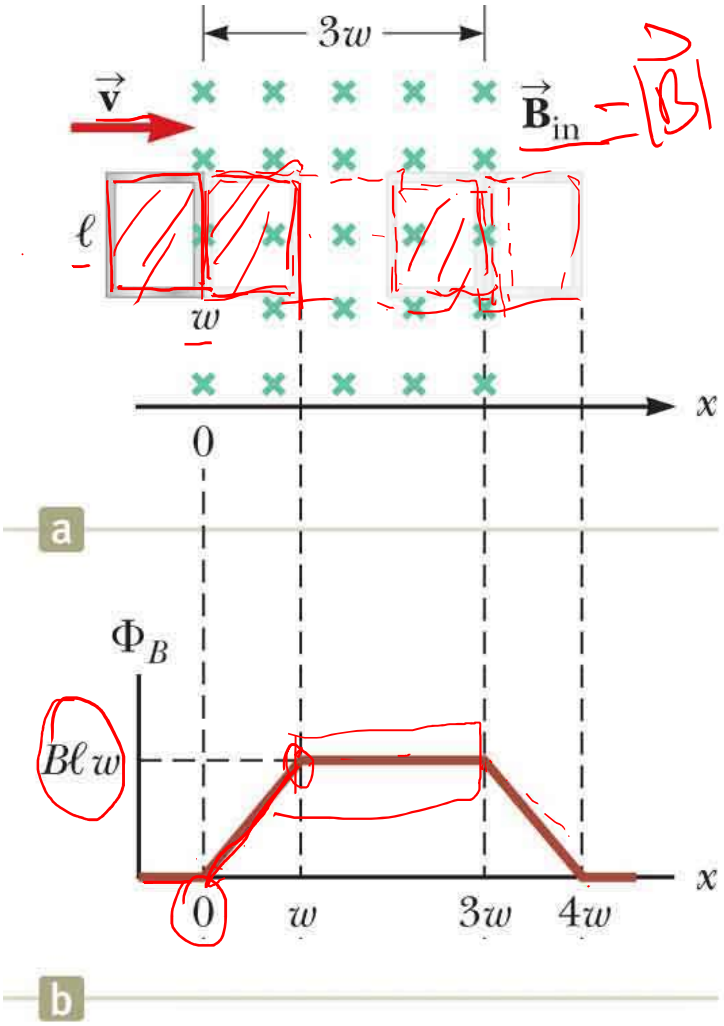
(B) Find the direction of the induced current in the loop when the magnet is pulled away from the loop.



Conceptual Example 30.5: A Loop Moving Through a Magnetic Field

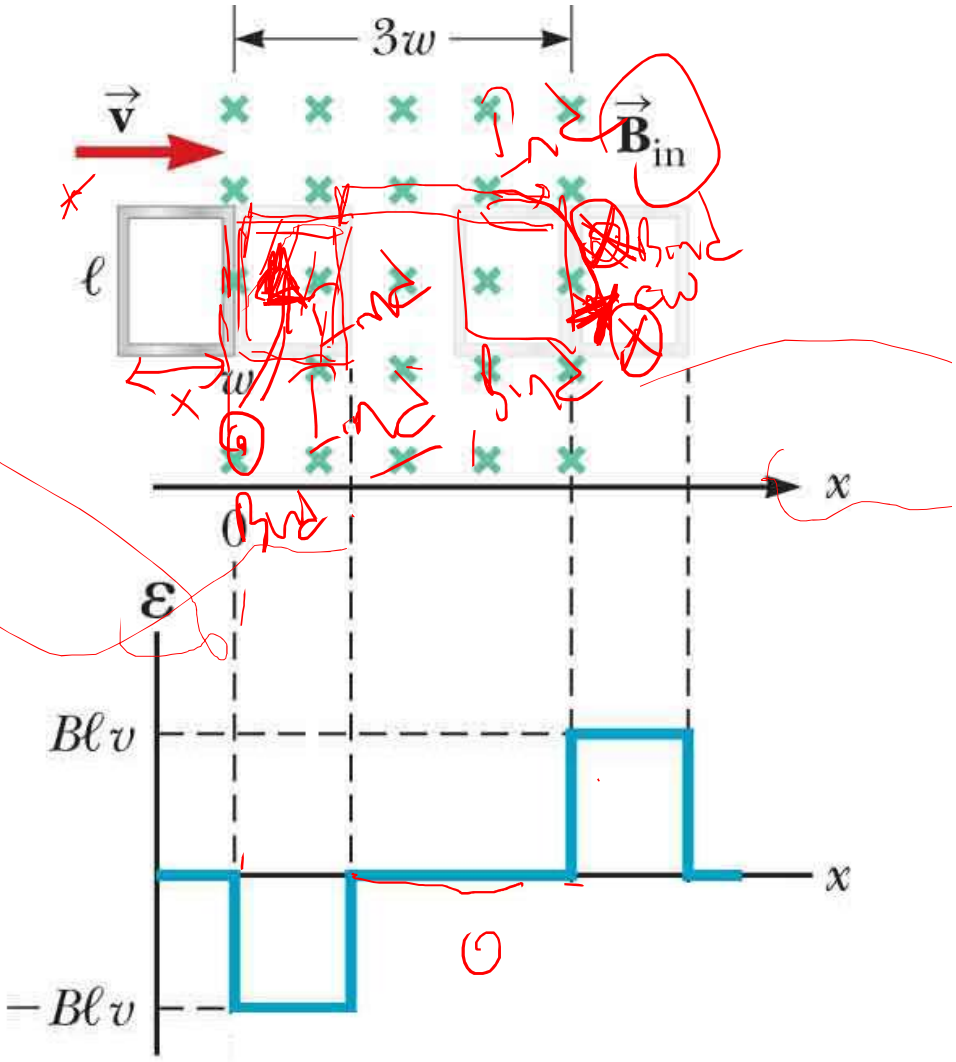
A rectangular metallic loop of dimensions ℓ and w and resistance R moves with constant speed v to the right as in the top figure. The loop passes through a uniform magnetic field $\vec{\mathbf{B}}$ directed into the page and extending a distance $3w$ along the x axis. Define x as the position of the right side of the loop along the x axis.

(A) Plot the magnetic flux through the area enclosed by the loop as a function of x.



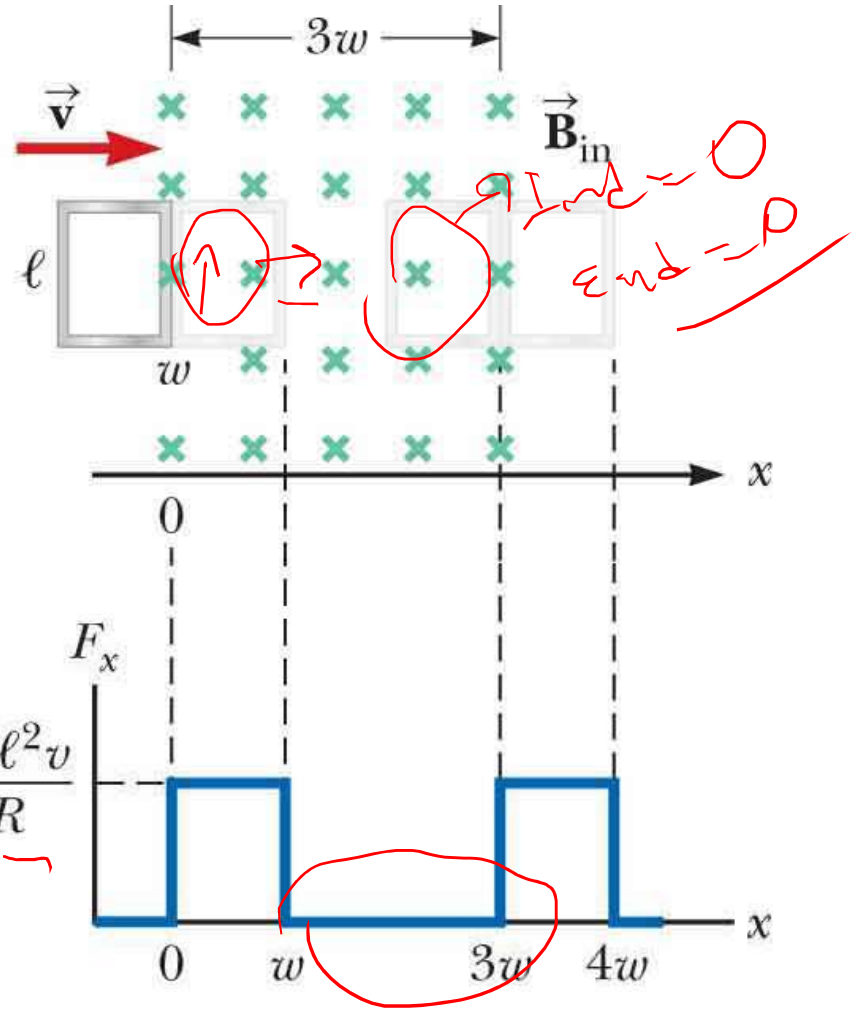
Conceptual Example 30.5: A Loop Moving Through a Magnetic Field

(B) Plot the induced motional emf in the loop as a function of x.



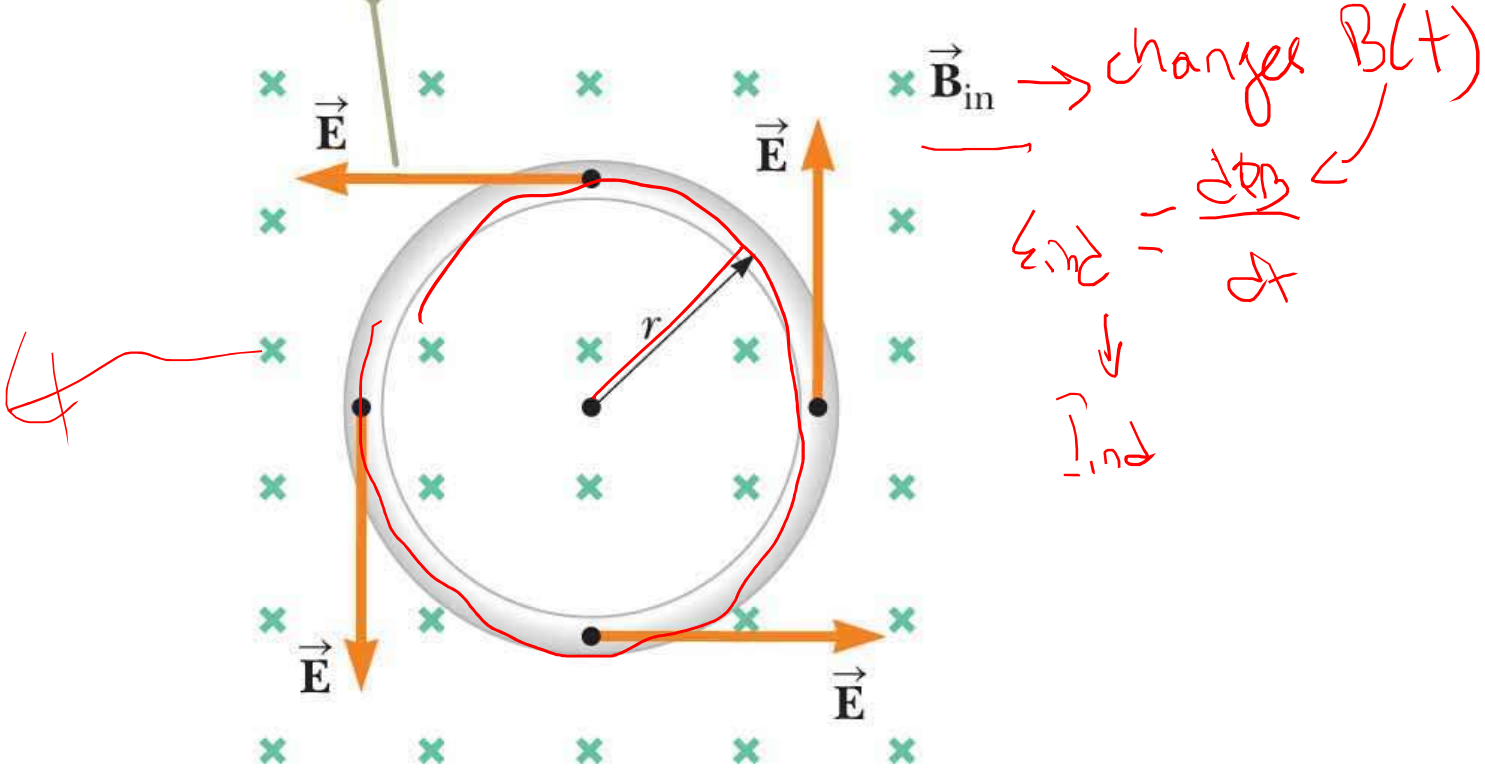
Conceptual Example 30.5: A Loop Moving Through a Magnetic Field

(C) Plot the external applied force necessary to counter the magnetic force and keep v constant as a function of x.



The General Form of Faraday's Law

If $\vec{\mathbf{B}}$ changes in time, an electric field is induced in a direction tangent to the circumference of the loop.



The General Form of Faraday's Law

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} = -\frac{d\Phi_B}{dt}$$

$$E \cdot \oint ds = -\frac{d}{dt}(BA)$$

$$\Rightarrow E(2\pi r) = -\frac{dB}{dt}(\pi r^2)$$

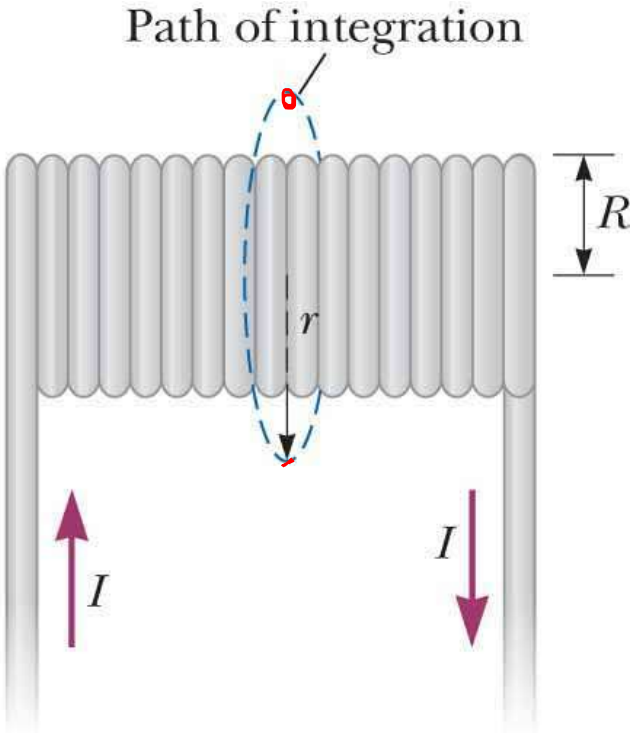
$$\Rightarrow E = -\frac{r}{2}\frac{dB}{dt}$$

$$\Delta V = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}}$$

Example 30.6: Electric Field Induced by a Charging Magnetic Field in a Solenoid

A long solenoid of radius R has n turns of wire per unit length and carries a time-varying current that varies sinusoidally as $I = I_{max} \cos \omega t$, where I_{max} is the maximum current and ω is the angular frequency of the alternating current source.

(A) Determine the magnitude of the induced electric field outside the solenoid at a distance $r > R$ from its long central axis.



Example 30.6: Electric Field Induced by a Charging Magnetic Field in a Solenoid

$$-\frac{d\Phi_B}{dt} = -\frac{d}{dt}\left(B\pi R^2\right) = -\pi R^2 \frac{dB}{dt}$$

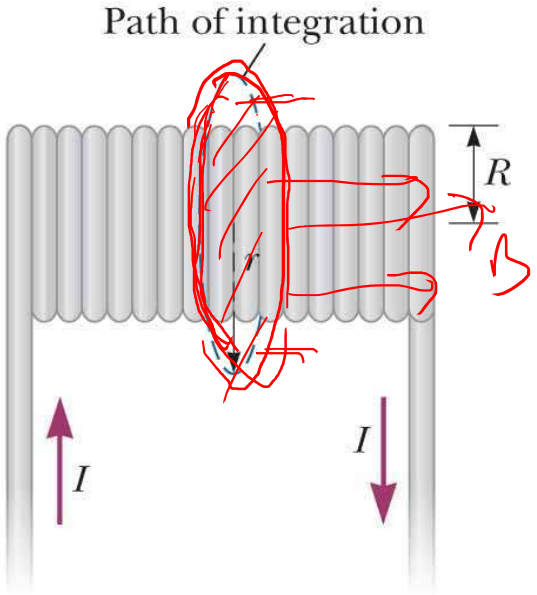
$$B = \mu_0 nI = \mu_0 nI_{\max} \cos\omega t$$

$$-\frac{d\Phi_B}{dt} = -\pi R^2 \mu_0 nI_{\max} \frac{d}{dt}\left(\cos\omega t\right) = \pi R^2 \mu_0 \omega \sin\omega t$$

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} = E\left(2\pi r\right)$$

$$E\left(2\pi r\right) = \pi R^2 \mu_0 nI_{\max} \omega \sin\omega t$$

$$E = \boxed{\frac{\mu_0 nI_{\max} \omega R^2}{2r} \sin\omega t} \quad \left(\text{for } r > R\right)$$



Example 30.6: Electric Field Induced by a Charging Magnetic Field in a Solenoid

(B) What is the magnitude of the induced electric field inside the solenoid, a distance r from its axis?

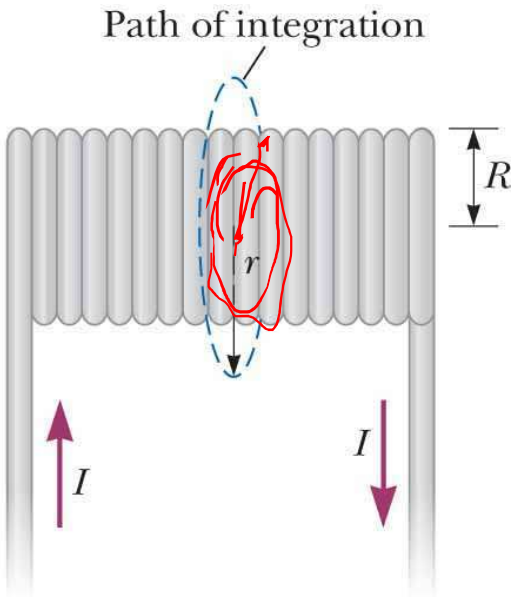
$$-\frac{d\Phi_B}{dt} = -\frac{d}{dt}\left(B\pi R^2\right) = -\pi R^2 \frac{dB}{dt}$$

$$-\frac{d\Phi_B}{dt} = -\pi r^2 \mu_0 n I_{\max} \frac{d}{dt}\left(\cos\omega t\right)$$

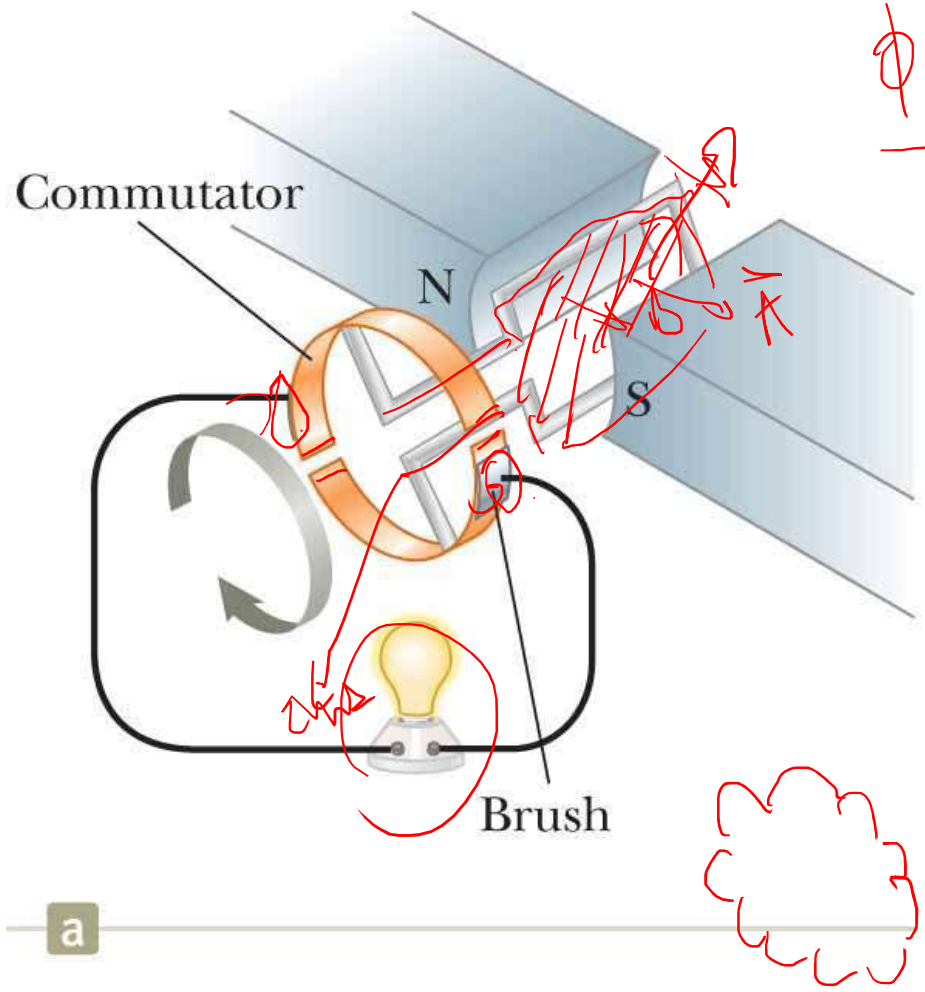
$$= \pi r^2 \mu_0 n I_{\max} \omega \sin \omega t$$

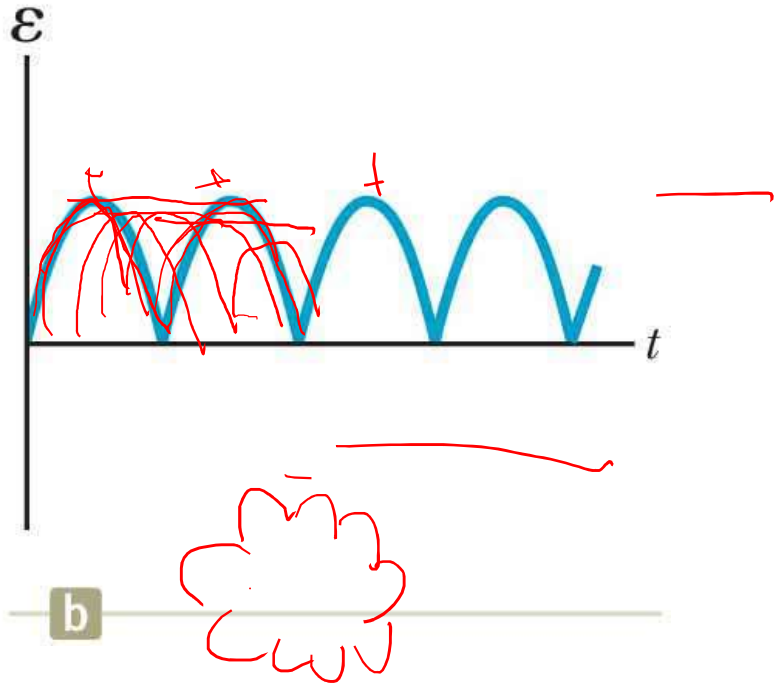
$$E\left(2\pi r\right) = \pi r^2 \mu_0 n I_{\max} \omega \sin \omega t$$

$$E = \boxed{\frac{\mu_0 n I_{\max} \omega}{2} r \sin \omega t} \quad \left(\text{for } r < R\right)$$

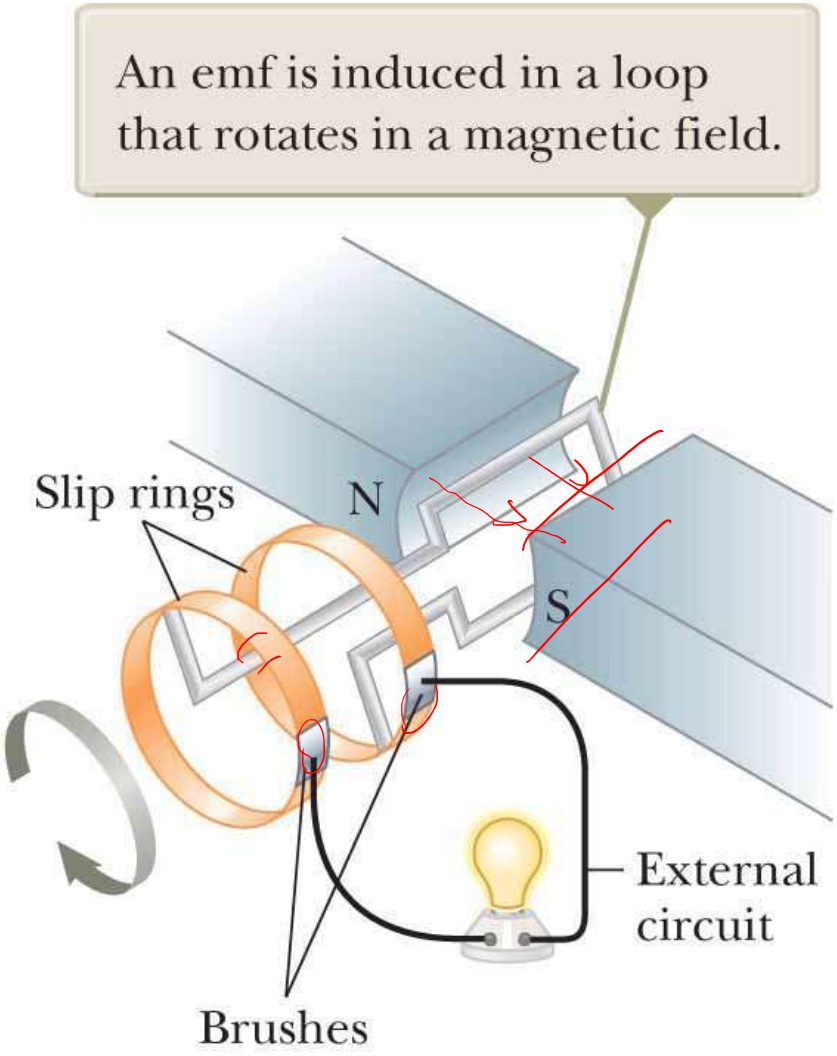


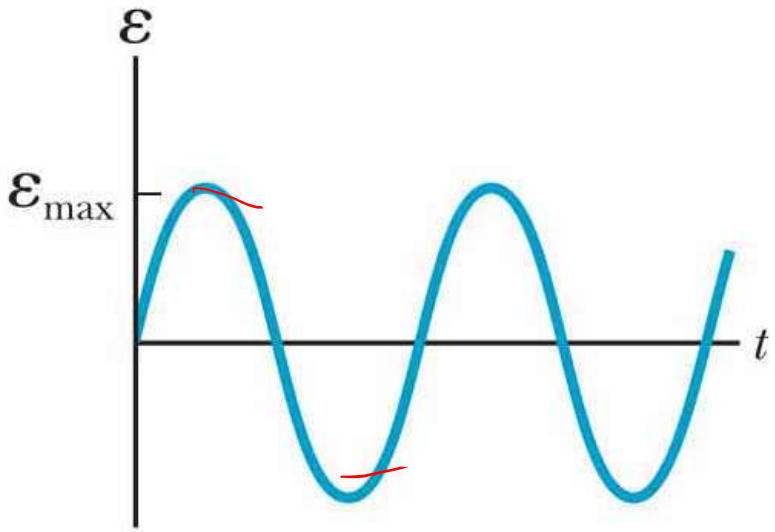
Direct-Current (DC) Generator





Alternating-Current (AC) Generator



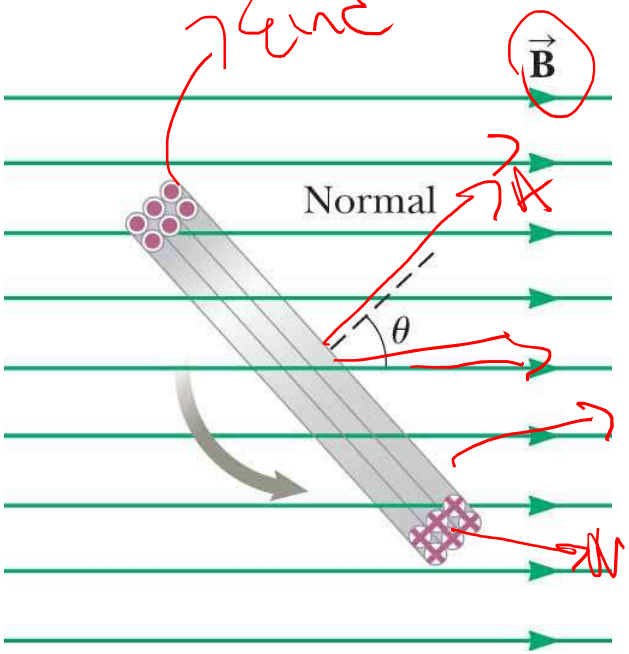


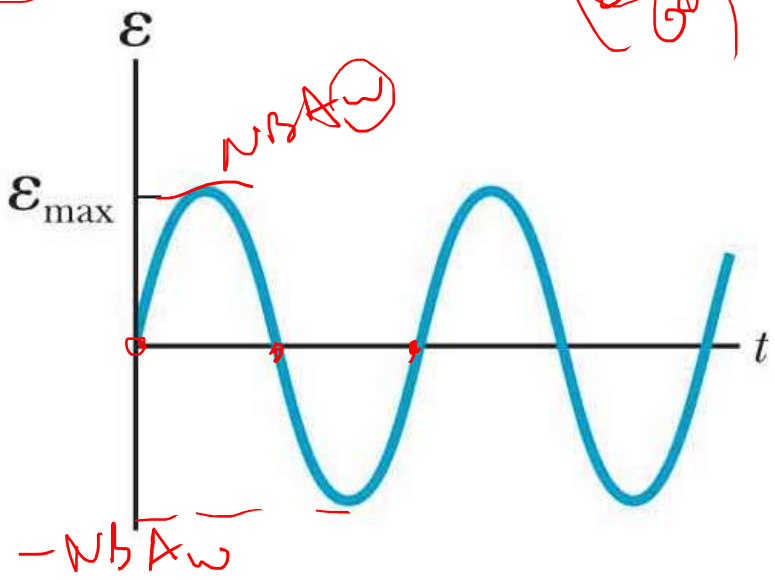
Sinusoidal Nature of Induced emf

$$\Phi_B = BA\cos\theta = BA\cos\omega t$$

$$\varepsilon = -N\frac{d\Phi_B}{dt} = -NBA\frac{d}{dt}(\cos\omega t) = NBA\omega\sin\omega t$$

$$\varepsilon_{\max} = NBA\omega$$





Quick Quiz 30.4

In an AC generator, a coil with N turns of wire spins in a magnetic field. Of the following choices, which does *not* cause an increase in the emf generated in the coil?

(a) replacing the coil wire with one of lower resistance

(b) spinning the coil faster

(c) increasing the magnetic field

(d) increasing the number of turns of wire on the coil

Quick Quiz 30.4

In an AC generator, a coil with *N* turns of wire spins in a magnetic field. Of the following choices, which does *not* cause an increase in the emf generated in the coil?

(a) replacing the coil wire with one of lower resistance

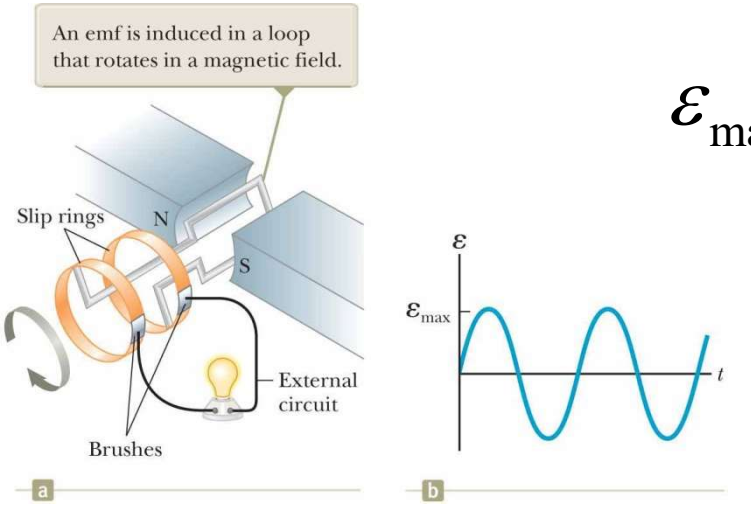
(b) spinning the coil faster

(c) increasing the magnetic field

(d) increasing the number of turns of wire on the coil

Example 30.7:
emf Induced in a Generator

The coil in an AC generator consists of 8 turns of wire, each of area A = 0.090 0 m^2, and the total resistance of the wire is 12.0 Ω. The coil rotates in a 0.500-T magnetic field at a constant frequency of 60.0 Hz.
(A) Find the maximum induced emf in the coil.

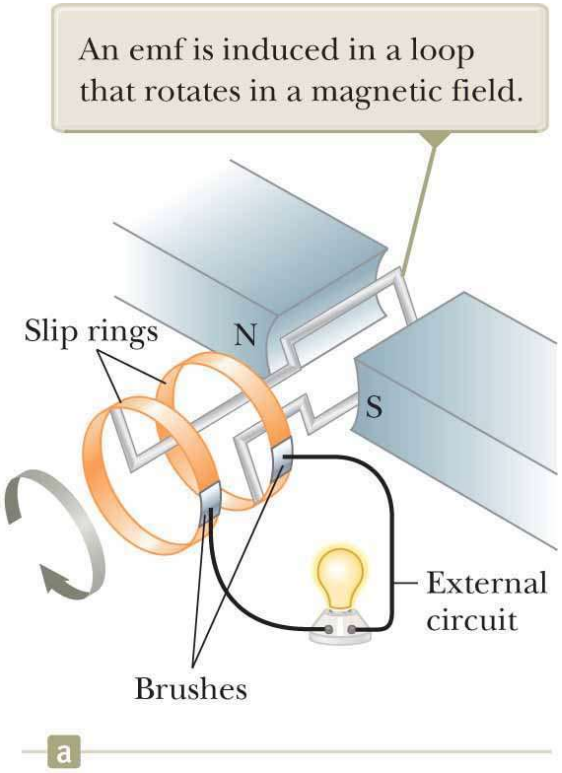


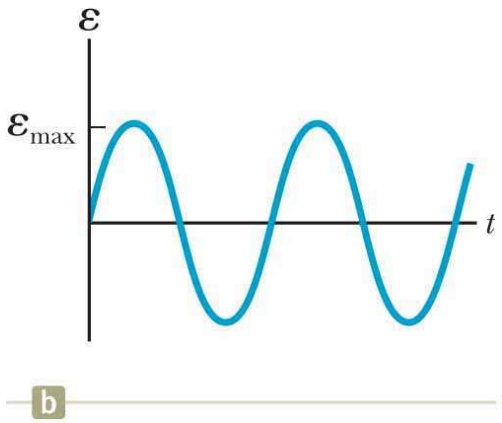
$$\varepsilon_{max} = NBA\omega$$
$$= NBA(2\pi f)$$

$$\varepsilon_{max} = 8(0.500\ \text{T})(0.0900\ \text{m}^2)(2\pi)(60.0\ \text{Hz}) = \boxed{136\ \text{V}}$$

Example 30.7: emf Induced in a Generator

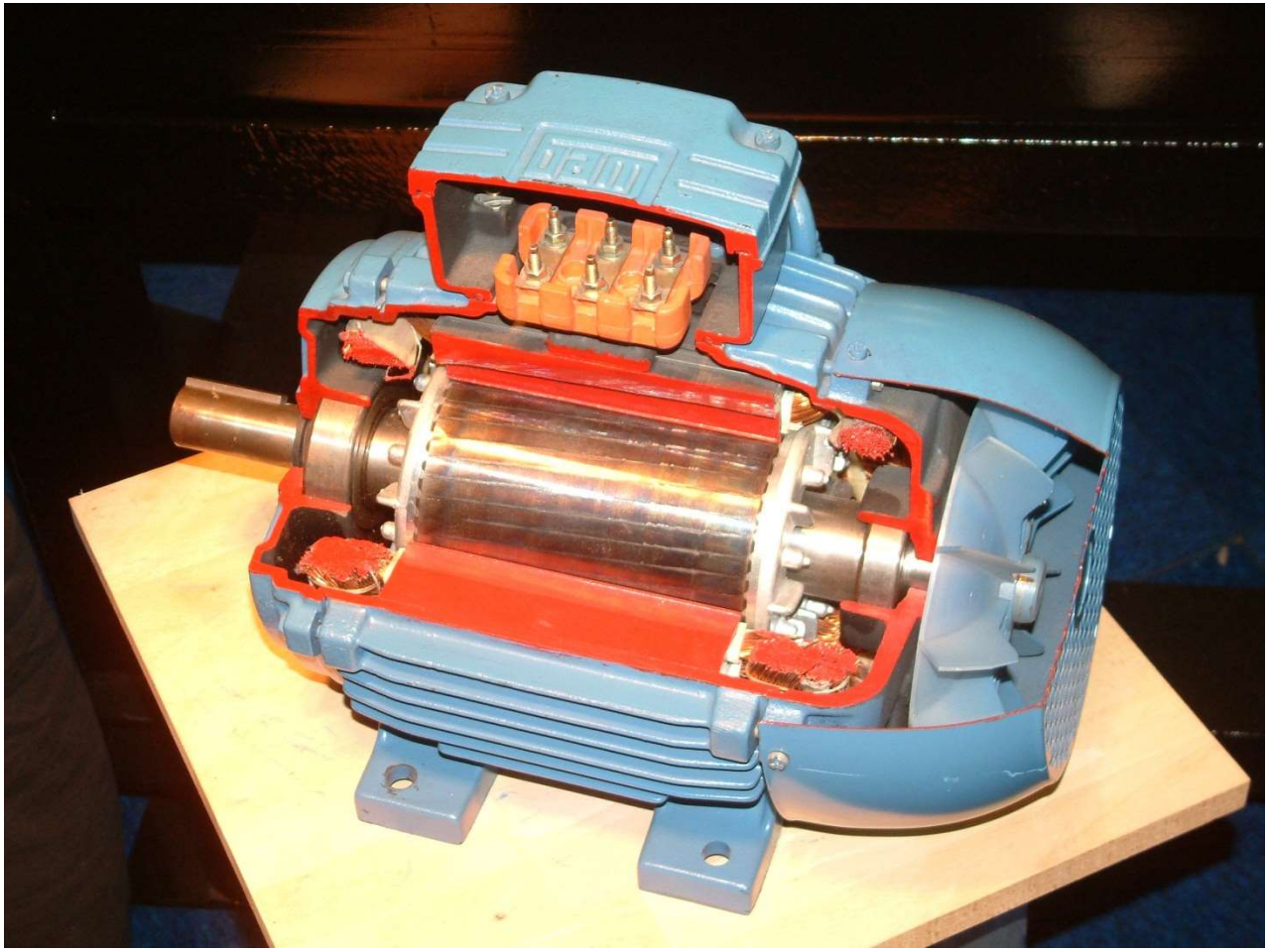
(B) What is the maximum induced current in the coil when the output terminals are connected to a low-resistance conductor?





$$I_{\max} = \frac{\mathcal{E}_{\max}}{R} = \frac{136\ \text{V}}{12.0\ \Omega} = \boxed{11.3\ \text{A}}$$

Motors



Example 30.8:
The Induced Current in a Motor

A motor contains a coil with a total resistance of 10 Ω and is supplied by a voltage of 120 V. When the motor is running at its maximum speed, the back emf is 70 V.
(A) Find the current in the coil at the instant the motor is turned on.

$$I = \frac{\mathcal{E}}{R} = \frac{120\ \text{V}}{10\ \Omega} = \boxed{12\ \text{A}}$$

Example 30.8:
The Induced Current in a Motor

(B) Find the current in the coil when the motor has reached maximum speed.

$$I = \frac{\mathcal{E} - \mathcal{E}_{\text{back}}}{R} = \frac{120\ \text{V} - 70\ \text{V}}{10\ \Omega} = \frac{50\ \text{V}}{10\ \Omega} = \boxed{5.0\ \text{A}}$$

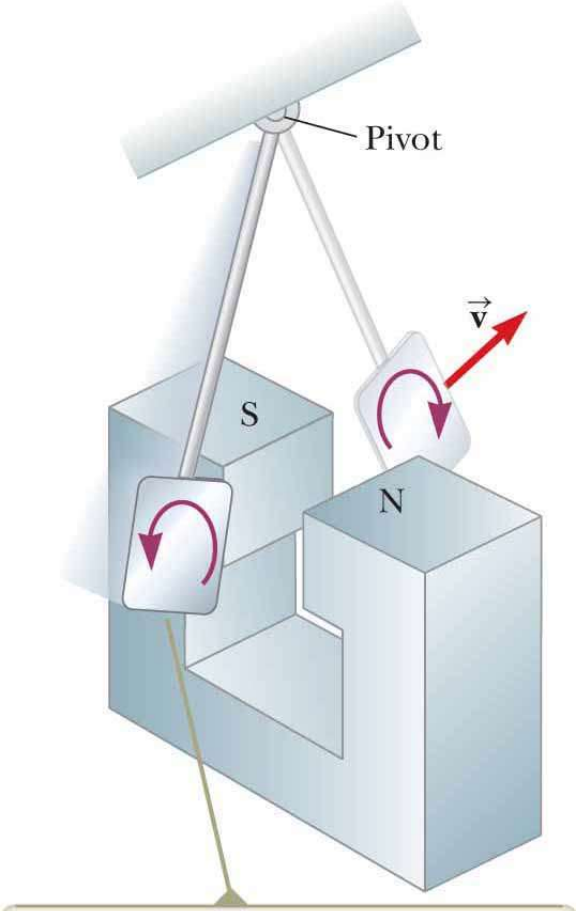
Example 30.8:
The Induced Current in a Motor

Suppose this motor is in a circular saw. When you are operating the saw, the blade becomes jammed in a piece of wood and the motor cannot turn. By what percentage does the power input to the motor increase when it is jammed?

$$\frac{P_{\text{jammed}}}{P_{\text{not jammed}}} = \frac{I_A^{\,2} R}{I_B^{\,2} R} = \frac{I_A^{\,2}}{I_B^{\,2}}$$

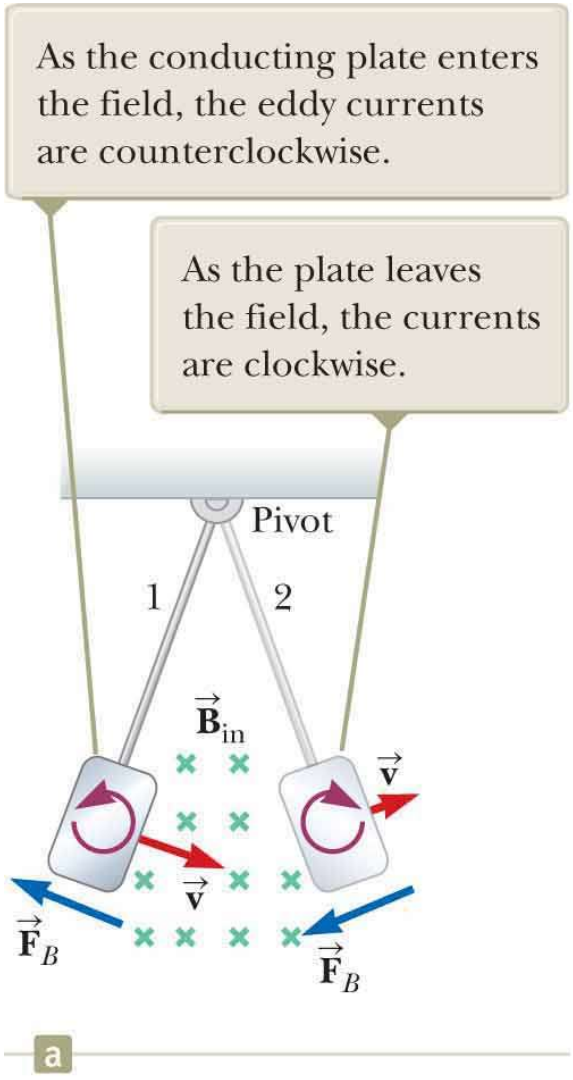
$$\frac{P_{\text{jammed}}}{P_{\text{not jammed}}} = \frac{(12\ \text{A})^2}{(5.0\ \text{A})^2} = 5.76$$

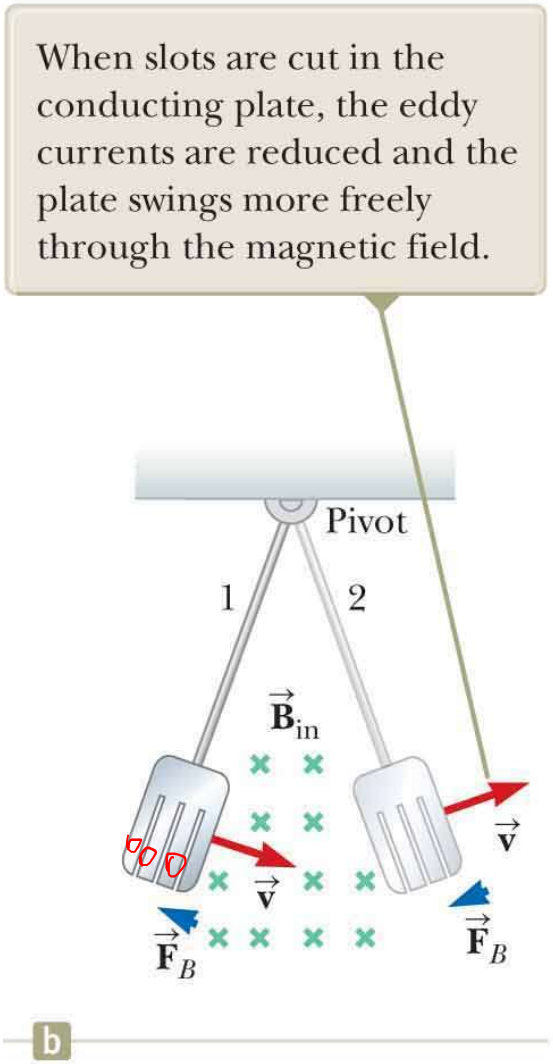
Eddy Currents



As the plate enters or leaves the field, the changing magnetic flux induces an emf, which causes eddy currents in the plate.

Eddy Currents





Assessing to Learn

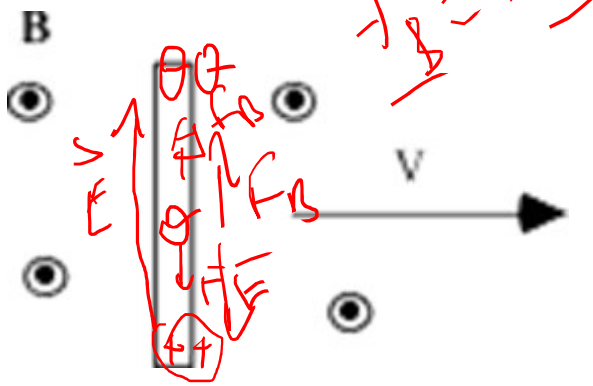
A long conducting bar moves with a constant velocity in a uniform magnetic field. The bar and its velocity are perpendicular to the magnetic field as shown. Which of the following statements are true?

A. At steady state there is an electric field in the bar.

B. At steady state there is a current in the bar.

C. At steady state there is a magnetic force on bar.

1. A only
2. B only
3. C only
4. A and B only
5. A and C only
6. B and C only
7. A, B, and C



Assessing to Learn

For which of the following is an emf induced?

A. A conducting rod is pulled on conducting rails that are placed in a uniform magnetic field directed into the page.

B. A conducting loop moves through a uniform magnetic field directed into the page.

C. A conducting loop rotates in a uniform magnetic field directed into the page.

D. A conducting loop moves in a magnetic field produced by an infinite current-carrying wire.

1. A only
2. A and B only
3. A and C only
4. A and D only
5. B and C only
6. A, B, and C
7. A, C, and D
8. All of them
9. None of the above

